Unconventional Monetary Policy and Deposit Retention

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Abstract

The two dimensions of unconventional monetary policy — holding the short rate at the zero-lower bound (ZLB) and quantitative easing (QE) — shifted deposit composition toward more interest-sensitive customers. The ZLB drew these customers by creating rate compression, which limited the benefit of searching for high-yielding liquidity sources. QE also drew rate-sensitive, large depositors from the Federal Reserve's purchases of bonds from non-banks. Once policy normalized, flow sensitivity and the deposit beta both increased with exposure to QE and the ZLB. We conclude that unconventional monetary policy made deposit retention costlier and riskier for banks.

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1 Introduction

Central banks initiated two unconventional monetary policies following the wreckage of the Global Financial Crisis (GFC). First, they committed to holding the short-term policy rate at its zero lower bound for an extended period (ZLB); second, they initiated Quantitative Easing (QE) to lower long-term rates directly. This paper shows that both dimensions of unconventional policies shifted the composition of bank deposits toward more rate-sensitive customers, making it costlier and riskier for banks to retain deposits once policy normalized.

The ZLB policy led to interest-rate spread compression, both across banks (because deposit rates collapsed toward zero) and between banks and money markets (because money market rates also collapsed toward zero). As a consequence, rate-sensitive liquidity demanders lost the incentive to move funds into high-rate substitutes for traditional bank deposits, or between banks based on differences in offered interest rates. This change both increased the flow of overall deposits into banks (see Figures 1 and 2) and also meant that allocation of deposits across banks lost its connection to deposit rates. As shown in Figure 3, the composition of bank deposits shifted away from interest-bearing time deposits and toward checkable, non-interest bearing accounts. This compositional change in deposits, as a consequence of the ZLB, reflects inflows from interest-sensitive customers, which emerge later when the short rate lifts off zero. We show that this change in composition affected how deposit supply responded when monetary conditions normalized.[1]

QE, or large-scale purchases by the Federal Reserve ("the Fed") of long-term bonds, also affected bank financing by mechanically increasing deposits when the bonds sold originate from outside the banking system (Hancock and Passmore, 2015; Christensen and Krogstrup, 2019, 2022; Acharya et al., 2024).[2] This component of unconventional

[1]Unlike the Fed, the European Central Bank set its policy rate below zero. This policy created additional funding problems for banks because they were unable to pass negative rates through fully to depositors. See Heider, Saidi and Schepens (2019) for a discussion of the policy tradeoffs.

[2]Deposits created by QE enter into banks, with the reserves created diluting the average risk and return of their asset portfolios. Banks thus rebalance themselves toward long-term bonds and other risky assets such as loans, and the portfolio rebalancing activity reaches the wide universe of U.S. banks.

monetary policy increased the share of uninsured deposits at banks, as we show, thus potentially harming funding stability. Uninsured depositors are both more interest-rate sensitive on average than insured ones, and they also respond to solvency shocks (because uninsured depositors are exposed to bank failure risk). This latter, latent risk became manifest during the SVB run and collapse (Jiang et al., 2024; Drechsler et al., 2026).

We estimate the effects on bank deposits of these two dimensions of unconventional policy simultaneously, an exercise challenged by the fact that QE, by design, overlaps with the ZLB. The two policies have different effects on bank stability — one potentially changes interest-rate sensitivity (the ZLB), while the other affects that and also run risk by increasing uninsured deposits (QE). But testing their effects separately poses a challenge because central banks resort to QE *only* when they have reached the limit of the standard monetary tool, the short-term money market rate, by putting that rate at its zero lower bound. Hence, the two policies are inherently coincidental.

We are able to separate the effects of the two policies by exploiting a new instrument which provides bank-market level variation in exposure to QE, following Yang (2026). The instrument uses granular data on Fed bond purchases, which are publicly disclosed at the transaction level. When these bond purchases originate from non-banks, they mechanically increase deposits at banks, but they are always intermediated at the original point of sale through a small number of large banks, the so-called Primary Dealers (PDs). Hence, the initial deposits created are not randomly distributed across the system. The instrument exploits the fact that non-PD banks with high market overlap with the PDs experience deposit inflows due to these Fed bond purchases.[3] And, the effect on deposits is strongest both statistically and economically for uninsured deposits. Thus, we can construct a time-varying measure of sensitivity to QE at both the bank and the bank-county levels.

Our core model relates deposit flows to deposit interest rates, to exposure to QE and

[3]This measure of QE exposure illustrates the Cantillon Effect, which argues that money creation has non-neutral effects because it does not reach all economic agents at the same time, as the metaphorical "helicopter drop" would.

its interaction with the deposit rate, and to exposure to ZLB and its interaction with the deposit rate. Following Begenau and Stafford (2022), we measure the deposit rate using their construct *Rate Gap*, which equals the difference between what a given bank pays for deposits compared to the average amount paid by competing banks raising deposits in the same local market(s). Rate Gap offers an exogenous (from the local market perspective) measure of a given bank's relative aggressiveness in pricing its deposits because multi-market banks engage almost completely in uniform pricing across local markets.[4] As such, a bank's relative deposit rate in a given market (compared to competing banks in that same market) isolates the supply side and does not reflect local funding demands (which in part drive local deposit growth apart from deposit rates). We first replicate Begenau and Stafford (2022), who show that Rate Gap strongly predicts deposit growth at the bank-county-year level (based on deposit growth in the Summary of Deposits); we then extend their results to show that averaging Rate Gap up to bank level (across local markets) provides a strong predictor of deposit growth at the bank-quarter level.

Our model shows that during the ZLB, Rate Gap's explanatory power for deposits falls sharply, especially at banks heavily exposed to QE. This happens because spreads compress toward zero (which is evident in the distribution of deposit rates as well as in Rate Gap — see Figures 4 and 5), which shifts the composition of deposits from interest-bearing accounts (especially time deposits) to non-interest bearing ones. Deposit growth thus loses its connection to interest rates during the ZLB, as differences in rates are just too small to matter. As such, deposits do not move across banks, and many new deposits flow into banks generally from money markets because banks provide marginally higher levels of protection and marginally higher levels of liquidity. Figure 1 shows that bank deposits grow rapidly relative to money market funds *during* the ZLB, and the reverse happens *outside* the ZLB. Said differently, without a pricing edge, money markets are at a severe competitive disadvantage relative to banks.

[4]For example, in 2019 the average US bank had over 100 branches but set just 1.1 unique interest rates across those branches; see Table 1 from Begenau and Stafford (2022).

We then test how deposit dynamics change during the two most recent rate cycles, one after the Global Financial Crisis (GFC) and the other after the Pandemic. Once interest rates lifted off the ZLB — first in late 2015 and again in 2022 — exposure to QE increased interest-rate sensitivity of deposit flows. That is, banks with high QE and ZLB exposure could only retain their deposits by increasing interest rates to match changes in market conditions. We further show that exposure to both QE and the ZLB increases bank deposit betas, which measure how aggressively banks raise deposit rates in response to increasing market rates. Unconventional policy shifted the composition of bank deposits toward interest-sensitive customers. As a result, when interest rates increased off the zero lower bound, banks with high exposure to those policies raised deposit rates more in order to help them retain those deposits, which otherwise would seek higher yield (either in the money markets or at other banks). Retaining deposits thus becomes more costly (i.e., higher deposit rates) and riskier (i.e., more rate sensitivity) due to banks' exposure to unconventional policy.

Our paper contributes to several strands of the emerging literature on bank funding during the post-GFC period.

First, the results speaks to the literature on the deposit franchise value. The seminal work by Drechsler, Savov and Schnabl (2017) on the deposits channel of monetary policy emphasizes the sensitivity of bank deposit supply to the spread between money market rates and bank deposit rates. Banks earn rents on sticky deposits — which generates the deposit franchise value — because this stickiness allows them to pay below-market rates. We show that rate sensitivity collapsed from unconventional monetary policy, leading to large deposit inflows. Despite these inflows, rate compression toward zero meant that the deposit franchise value was reduced by unconventional monetary policy (at least on a per-dollar basis). Yet when monetary conditions normalized, exposure to deposits from unconventional policy eroded the natural interest-rate risk hedge to offset asset-side losses with value generated by pricing deposits below market rates.[5]

[5] Drechsler, Savov and Schnabl (2021) report evidence from this natural hedge, but they do not consider the two most recent interest rate upcycles where this hedge becomes less effective. Their analysis ends in 2017.

In other words, banks heavily exposed to QE and to the ZLB had to raise deposit rates in order to keep deposits from flowing out. Consistent with our arguments, Usenko (2026) shows that interest rate sensitivity is highest among large depositors, exactly those which we show are drawn into the banking system by unconventional monetary policies.

Second, we extend Acharya et al. (2024), who argue that QE harmed bank stability by increasing their exposure to liquidity risk, both in lending and in taking deposits. They emphasize the asymmetry in liquidity exposure, whereby banks increase liquidity risk during QE episodes (which is offset by the associated increase in reserves), but do not lower this exposure when QE ends; this asymmetry results in more fragile banks as a legacy of the QE programs. Our mechanism focuses on how unconventional policy — *both* QE and the ZLB — affected deposit interest sensitivity during rate increasing periods, which we show increases at banks most exposed to the two components of unconventional policy.

Third, we contribute to the literature on fragility stemming from the Silicon Valley Bank (SVB) crisis.[6] This literature focuses mainly on SVB's pathological combination of long-term assets financed with almost 100 percent uninsured deposits. Jiang et al. (2024) estimate large duration-related losses at SVB (like many banks) and show that they had especially high levels of uninsured deposits, which increased fragility. Drechsler et al. (2026) argue that the deposit franchise itself is fragile because it can support two equilibria, one in which deposits stay in the bank and the bank earns rents on them when market interest rates rise (because spreads rise), thus offsetting asset-side losses; another equilibrium exists, however, in which depositors run under the rational belief that the bank will become insolvent if other depositors leave. This mechanism neatly explains the funding problem related to uninsured deposit fragility, but it does not address the broader effect of deposit composition on interest rate sensitivity, which is the focus of our paper. Blickle et al. (2025) is closest to our paper, as it shows higher deposit

[6] Cookson et al. (2026) emphasize the role of social media in amplifying and extending the effects of the SVB run.

flightiness after unconventional policy. However, our paper, unlike theirs, exploits cross-bank heterogeneity in exposure to both dimensions of unconventional policy. As such, we can separate the effects of the two components of that policy: the extended ZLB and large-scale asset purchases from QE.

2 Unconventional Monetary Policy

Monetary stimulus coming out of the Global Financial Crisis, based on reductions in the short-term interest rate, was viewed as insufficiently powerful to stimulate the economy and foreclose the possibility of a deflationary spiral like what had occurred during the 1930s (Friedman and Schwartz, 1963; Bernanke, 1983). In part the problem emerged from prior declines in interest rates, which had trended down since the middle of the 1980s and thus reduced the policy space for the Fed to lower interest rates (Hillenbrand, 2025). As a result, the Federal Reserve dropped its policy rate to near zero in late 2008, and added two new tools to amplify its influence on overall financial conditions. First, Forward Guidance was implemented by communicating the Fed's intent to leave the short rate at its "Zero Lower Bound" (ZLB) for an extended period of time. Second, the Fed initiated large-scale asset purchases, aka Quantitative Easing (QE), designed to put additional downward pressure on long-term interest rates by buying long-term Treasuries and Mortgage-Backed Securities. The Fed reinstituted both of these policies in response to the Pandemic.

Bernanke (2017) and Kuttner (2018) survey the policy rationales as well as the empirical evidence about the effectiveness of these unconventional tools. A number of studies find loosening of financial conditions such as declines in term premia around policy announcements (e.g., Gagnon et al., 2011; Krishnamurthy et al., 2011; Ehlers, 2012; Bauer and Neely, 2014) and increases in bank lending for more-exposed banks (Rodnyansky and Darmouni, 2017; Luck and Zimmermann, 2020; Chakraborty, Goldstein and MacKinlay, 2020).[7] Neither survey paper, however, considers how these policies could affect banking stability when monetary conditions normalized.

[7] Diamond, Jiang and Ma (2024), in contrast, argue that QE constrained lending by filling up bank balance sheets with reserves and thus displacing balance sheet capacity for lending.

We argue that both the persistent use of the ZLB as well as QE could increase bank exposure to interest-sensitive deposits. Prior to this period, bank deposits had been largely supplied by sticky customers with low interest elasticity.[8] As a result, deposit spreads increased when interest rates rose; although some deposits did flow out, the wider spread provided banks with a hedge against losses from holding interest-sensitive (i.e., long-term) assets like mortgages or mortgage-backed securities. The utility of this natural hedge could decline with both the persistent ZLB and by QE because both policies may change the interest sensitivity of bank deposits. Importantly, this sensitivity only becomes evident when unconventional policy ends and monetary conditions normalize.

The ZLB changes the composition of bank deposits because it leads to spread compression. As illustrated in Figures 4 and 5, the variation across banks in deposit rates declines sharply during these periods. Moreover, the difference in rates paid by the median bank versus that paid by money markets also declines (Figure 6). Hence, interest-sensitive liquidity demanders, for example corporate CFOs, lose the incentive to search for yield on their short-term assets. Prior to the ZLB period, many such agents would park cash in money markets rather than banks to receive higher returns (recall Figure 1).[9] With spread compression, bank deposits become a better option given the stronger explicit and implicit insurance against losses. This helps explain the rapid growth of deposits during the years following the GFC, when deposits grew about twice as fast as nominal GDP (recall Figure 2).

QE as envisioned by Bernanke was tantamount to a swap between private banks and the Fed of long-term assets for short-term bank reserves. In its basic form, the Fed would simply purchase a bond from a bank, and pay for that by issuing new reserves into the selling bank's Fed account. As such, the selling bank's liabilities are unaffected, while its portfolio of assets becomes shorter in duration and higher in liquidity. The

[8] Argyle et al. (2025) show that high-balance deposits exhibit higher stickiness; Egan et al. (2025) also show that depositor stickiness varies with deposit size and depositor demographics (age).

[9] The model crashed badly during the week following Lehman Brothers, leading to a government bailout of the money market industry.

policy effect works by moving duration risk from the private banking system to the government.

The problem with the textbook model of QE, where the Fed's actions only affect banks, is that most of the long-term bonds purchased originated *outside* the banking system. Carpenter et al. (2015) show, for example, that most of the bonds came from households (which is a catch-all category from the US Flow of Funds which includes hedge funds), insurance companies, pension funds, and brokerage firms. None of these entities can accept reserves from the Fed as payments for bonds sold. Instead, they are paid in deposits by one of the Primary Dealer banks, which interact directly with the Fed and act as an intermediary between the original seller and the Fed. The PD banks then resell the bond to the Fed in return for reserves. The deposits (and reserves) thus start at one of the PDs but then move to other banks. The new money created by QE propagates across the banking system gradually, with its largest immediate effects at banks located near the PD bank locations. Hence, QE in practice is not a simple asset swap between banks and the Fed; instead, it leads to an expansion of both the asset side (i.e., more reserves) and the liability side (more deposits) of banks. These new deposits, as we will show, are not distributed evenly across banks and do not behave similarly to normal deposits created by private market forces.

3 Data

This section details the data sources used in this paper. Our analysis draws on four sources. Bank-level financial information comes from the Federal Financial Institutions Examination Council (FFIEC) Call Reports; branch-level deposit information comes from the Federal Deposit Insurance Corporation (FDIC) Summary of Deposits (SOD) Data; transaction-level data on the Federal Reserve's open market operations, used to construct our QE deposit-channel instrument, come from the Federal Reserve Bank of New York's Open Market Operations transaction data; and aggregate macroeconomic time series come from Federal Reserve Economic Data (FRED). The sample spans 2001 through 2025, covering both the post-financial-crisis and the post-pandemic

zero-lower-bound episodes and the tightening cycles that followed each. Detailed variable definitions and summary statistics are provided in Appendix Tables A.1 and A.2, respectively.

3.1 FFIEC Call Report Data

Our bank-level analysis uses FFIEC Call Report data accessed through Wharton Research Data Services (WRDS). This dataset provides comprehensive accounting information from the Consolidated Reports of Condition and Income, which all insured commercial banks are required to file each quarter. For each bank-quarter, we retain only the most recent submission to incorporate amended filings. Because flow variables in the Call Reports — such as interest expense on deposits — are reported on a year-to-date basis, we transform them into discrete quarterly flows by subtracting the value from the preceding quarter within the same fiscal year. From these records, we construct each bank's deposits, insured and uninsured deposits, total assets, and its deposit rate, measured as annualized interest expense on deposits scaled by average deposits. This yields a consistent quarterly panel of bank balance-sheet and deposit-pricing outcomes.[10]

3.2 FDIC Summary of Deposits (SOD) Data

For our local, bank-county analysis we use the FDIC Summary of Deposits (SOD) data, which provide branch-level deposits and geographic identifiers for all FDIC-insured institutions. These data are released annually with a reporting date of June 30. We use them to construct, for each bank, the geographic distribution of its deposits across counties and its deposit market share within each county. Aggregating branch deposits to the bank-county-year level gives the local deposit measures used throughout the paper and the predetermined deposit-market weights that underlie our bank-county analysis.

[10]When a bank is involved in a merger or failure, we drop the bank-time or bank-county-time observation which contains the structural change, as this often has a massive effect on rates of change.

3.3 FRED Data

We supplement our analysis with aggregate macroeconomic time series from FRED to characterize the broader economic environment across monetary policy phases. These series include total deposits at commercial banks, money market mutual fund assets, nominal GDP, the Federal Funds rate, and the 3-month Treasury bill rate. We use these data, along with the Federal Funds rate path, to construct our descriptive figures and date zero-lower-bound episodes.

3.4 Federal Reserve Bond Purchase Data

To construct our core measure of QE's effect on bank deposits, we use transaction-level security trade data from the Federal Reserve Bank of New York's Open Market Trading Desk. These data record every asset purchase and sale conducted with primary dealers — the financial institutions authorized to transact directly with the Fed — and include the trade and settlement dates, transaction type, par and dollar amounts, CUSIP, and counterparty. Because the Fed's large-scale asset purchases flowed through these primary dealers, these records capture the initial pathway by which QE-created reserves and deposits entered the banking system. Our data span U.S. Treasuries, agency MBS, and CMBS across QE I–III and the pandemic-era expansion. Due to the Fed's two-year disclosure lag, our transaction sample extends through 2023Q4. To align this series with our overall sample period (2001–2025), we code quarters without transaction data as zero, reflecting the absence of QE operations prior to 2009 and during 2024–2025. Finally, we aggregate these net transaction values to the dealer-quarter and dealer-year levels to measure each dealer's net liquidity injection, which serves as the time-series input for our deposit-channel instrument.

4 Empirical Strategy

We report the effects of unconventional monetary policy on bank deposit growth in three steps. The first step establishes the relevance for deposit growth of our bank-level and bank-county level measures of exposure to QE (and we compare our measure with two other exposure measures from the literature). Second, we estimate how

accumulated QE exposure and the ZLB affected the sensitivity of deposit supply to deposit interest rates. To assess the effect of the ZLB, we simply incorporate an indicator variable interaction allowing rate sensitivity to differ during the quarters of the two ZLB regimes (one following the GFC and the other after the Pandemic). To assess QE, we interact deposit rates with the bank's total exposure to QE during the prior three years. In the third step, we test whether the accumulated effect of these two policies made deposit supply *more* interest-rate sensitive during the two interest rate cycles following the GFC. In this analysis, we focus on changes in deposit funding and rate sensitivity during the two rate cycles, each of which followed a period of unconventional policy. We test how deposit rate-sensitivity interacts with two cross-bank measures of policy exposure accumulated during the prior period. For the 2016 Q1-2020 Q1 interest rate cycle, QE and ZLB exposures reflect backward-looking moving averages from the period from 2009-2015; for the 2022 Q2-2025 Q4 interest rate cycle, the two exposures reflect the period from 2020-2021.

4.1 QE and Bank Deposit Growth

Tables 1 and 2 report the contemporaneous effect of bank (bank-county) QE exposure on deposit growth. The two strategies have different strengths. The Call Report data, which is bank-level, allows us to exploit higher-frequency variation (quarter v. year); it also allows us to separate deposits into insured vs. uninsured. The Summary of Deposits data, in contrast, allows us to measure deposit growth at bank-county level, but only at yearly frequency and without a way to decompose deposits by type.

Table 1 reports estimates at the bank-quarter level, using the following specification:

$$\Delta \text{Dep}^{k}_{b,t} / \text{Assets}_{b,t-1} = \beta_1 \, Z^{R1}_{b,t} + \beta_2 \, Z_{\text{mbs},b,t} / \text{Assets}_{b,t-1} + \beta_3 \, Z_{b,t} / \text{Assets}_{b,t-1} + \alpha_b + \gamma_t + \epsilon_{b,t} \tag{1}$$

where $\Delta \text{Dep}^{k}_{b,t}$ denotes the quarterly change in deposits of category k for bank b, with k indexing total, uninsured, and insured deposits across Panels A, B, and C, respectively. Each dependent variable is scaled by lagged total assets.

The three regressors in Equation (1) represent alternative measures of contempo-

raneous QE exposure. $Z_{b,t}^{R1}$ is a bank-level reserve instrument based on Acharya et al. (2024); this variable equals a bank's expected growth in reserves using the product of its past reserve usage (the 4-quarter moving average of the bank's lagged share of total reserves), multiplied by the growth rate of aggregate reserves. $Z_{\text{mbs},b,t}$ is also a shift-share instrument capturing the direct asset-swap effect of the Fed's mortgage-backed security component of QE; the measure, inspired by Rodnyansky and Darmouni (2017), equals the product of each bank's share of outstanding mortgage-backed securities times the amount of MBS bonds purchased by the Federal Reserve.[11]

$Z_{b,t}$ is the deposit channel instrument developed first in Yang (2026). The variable equals a deposit-weighted average of county-level QE exposure based on the amount of bonds intermediated through each Primary Dealer (PD). Bond purchases are allocated to each county based on each dealer's share of its business in the locality. The deposit weights depend on pre-determined, bank-county deposits from SOD. Variation in QE exposure at the county-time level occurs from four sources: 1) changes in the amount of bonds purchased by the Fed; 2) variation in the amount of those bonds intermediated by each PD; 3) variation in each PD's share of its business in each locality; and 4) variation in each bank's market share in each county.[12] The idea is that bonds intermediated through a given PD are likely to originate in areas where that PD has a market presence (because its customers are there), and that the deposits created by the transaction will flow into other nearby non-PD banks, either because the bond seller has an account there or due to broad business connections between the PD and other banks operating in the same areas. In all of our tests that follow, we leave out the PD banks themselves.

Both $Z_{\text{mbs},b,t}$ and $Z_{b,t}$ are expressed in dollars and scaled by the prior quarter's total bank assets for comparability. The $Z_{b,t}^{R1}$ instrument is constructed as a growth rate. To account for unobserved heterogeneity, we include bank fixed effects (α_b) and quarter

[11] Rodnyansky and Darmouni (2017) use banks' MBS holdings as a percentage of their assets to capture exposure in a difference-in-differences research design whereby the balance sheet ratio defines treatment exposure to various episodes of QE. They show that banks with more MBS increased lending during the first and third QE episodes; in contrast, they find no effect of Treasury Bond exposure on lending. Rather than using indicator variables turned on during various periods of Fed bond buying, our approach embeds the actual amount of MBS bonds purchased by the Fed in each quarter.

[12] Appendix A.1 provides a full explanation of the construction of this instrument.

fixed effects (γ_t), and two-way cluster standard errors at the bank and quarter levels.[13]

We estimate Equation (1) over four sample periods corresponding to distinct phases of the Federal Reserve's balance-sheet policy: the full sample (2009-2025), the QE I–III era (2009-2014), the Post-QE and Quantitative Tightening period (2015-2019), and the Pandemic QE era (2020-2021). Within each period, we report two specifications: the first includes only the MBS and deposit channel measures, and the second adds the reserve instrument. Because a reserve-based measure can be constructed only for the subset of large banks that report reserves in their Call Reports, the specifications that include $Z_{b,t}^{R1}$ are estimated on a smaller sample; presenting the two side by side isolates the role of the reserve measure while holding the remaining structure fixed. Splitting the sample by policy phase allows the relationship between each exposure measure and deposit flows to differ across periods of balance-sheet expansion, tightening, and renewed expansion, rather than constraining it to a single average effect over a period spanning very different monetary regimes.

[Insert Table 1 Here]

Table 1 shows that only the deposit channel instrument ($Z_{b,t}$) consistently covaries positively with bank-quarter deposit growth. We find this measure of QE exposure is associated with faster deposit growth, and the magnitude is driven by the periods when QE was active: 2009 to 2014 (columns 3 and 4), and 2020 to 2021 (column 7 and 8). The other two measures — one based on reserve growth and the other on growth in mortgage-backed securities — have either weak correlations statistically, or inconsistent sign patterns, or both. We think this makes sense because each of these measures captures asset-side exposure to QE; QE affects bank reserves directly when the Federal Reserve purchases assets from banks, and banks holding more MBS can sell directly to the Fed. But such transactions are asset swaps which do not change deposits. In contrast, deposits increase mechanically when the Fed purchases bonds that originate

[13] Summary statistics for the three instruments and the other variables are reported in Appendix Table A.2.

outside the banking system because the bond seller can only be compensated for the sale with deposits (only banks can hold reserves directly). The ($Z_{b,t}$) instrument captures this channel because it allocates all of the bonds that the Fed purchases across banks based on where each bank is raising its deposits (as opposed to variation in each bank's asset holdings).

Panels B and C of Table 1 report Equation (1) after dividing the dependent variable into the change in uninsured and change in insured deposits, again each scaled by lagged assets. These sum to the change in total deposits, so the regression coefficients in Panels B and C sum to what is reported in Panel A. This decomposition of the dependent variable shows that the effect of QE on deposit growth is concentrated in uninsured deposits, as the magnitude of the coefficient on $Z_{b,t}$ is twice as large during the first rounds of QE and three or mores as large during the second, compared to insured deposits. And the effect of $Z_{b,t}$ is not statistically significant for insured deposit growth in the full sample. Interestingly, we find a negative correlation between the reserve-based measure ($Z^{R1}_{b,t}$) and insured deposits.

Table 2 reports similar deposit-growth regression at the bank-county-year level, from the following specification:

$$\Delta \text{Dep}_{b,c,t} / \text{Assets}_{b,t-1} = \beta \, Z_{b,c,t} / \text{Assets}_{b,t-1} + \theta_{b,t} + \lambda_{c,t} + \eta_{b,c} + \epsilon_{b,c,t} \tag{2}$$

where $\Delta \text{Dep}_{b,c,t}$ denotes the annual change in total deposits for bank b in county c from SOD year $t-1$ to t, and $Z_{b,c,t}$ is the bank-county-level deposit channel instrument constructed in Yang (2026). This more granular approach captures the fact that deposits at a given bank may increase more in counties where primary dealer banks are actively purchasing bonds.[14]

Both the dependent variable and the instrument, as before, are scaled by the prior year's total bank-level assets.[15] To account for unobserved heterogeneity, we include

[14]The version applied above in Equation (1) comes from the aggregation of the bank-county measures up to bank level.

[15]Each is winsorized at the 1st and 99th percentiles.

bank×year fixed effects ($\theta_{b,t}$) and county×year fixed effects ($\lambda_{c,t}$), with additional bank×county fixed effects ($\eta_{b,c}$) in the full-sample specifications.[16] The county×year fixed effects absorb all variation common to a local market in a given year — including local credit demand, income shocks, and market-wide flight-to-safety flows — so that identification comes from comparing banks with different deposit channel exposure within the same county and year. The bank×year fixed effects remove deposit demand, as well as any other bank-time level characteristic. These fixed effects also absorb both of the asset-swap measures of QE exposure ($Z^{R1}_{b,t}$ and $Z_{\text{mbs},b,t}$). The model is estimated at an annual frequency, consistent with the reporting cycle of the FDIC Summary of Deposits (SOD) data; year t spans July 1 of year $t-1$ to June 30 of year t, aligning the deposit inflows measured in $Z_{b,c,t}$ with the period over which the deposit changes are recorded. We cluster standard errors two-way at the bank and county levels.

We estimate Equation (2) over the full sample (2009–2025) and two sub-periods corresponding to distinct phases of Federal Reserve balance-sheet policy: the QE I–III era (2009–2014) and the Pandemic QE era (2020–2021). Column (1) reports the full sample; Column (2) restricts it to small banks, defined as those with total assets below $10 billion as of year-end 2008; and Columns (3) and (4) report the QE I–III and Pandemic QE sub-periods. The small-bank column assesses whether the deposit channel operates across the broader local banking market rather than only at the largest institutions, while the sub-period splits allow the instrument's predictive power to differ across phases of balance-sheet expansion and to scale with the intensity of Federal Reserve purchases, rather than being constrained to a single average over regimes with very different purchase volumes.

These results provide powerful support that exposure to QE not only increases deposits but does so with heterogeneous effects across markets. $Z_{b,c,t}$ incorporates the idea that when a bond is purchased outside the banking system, the deposits generated by that purchase are likely to end up in banks with high market overlap with the

[16] Including the bank-county effect means that identification requires bank-county time variation; since such variation is limited for the sub-period models, we leave this fixed effect out.

primary dealer banks. The magnitudes suggest that a one-standard-deviation increase in QE exposure increases deposits by about 0.13% ($= 0.25 \times 0.005$) of assets during the first wave (2010-2014), and almost 1% ($= 1.92 \times 0.005$) of assets during the second (2020-2021).

[Insert Table 2 Here]

4.2 Deposit-Rate Sensitivity and Unconventional Policy

As we have argued, both dimensions of unconventional monetary policy potentially shift the composition of deposits. The ZLB-induced compression in rates paid both across banks and between banks and money markets, which in turn may have led normally interest-sensitive agents to park liquidity in banks as long as rates remained severely depressed by policy. And, as we have shown, QE increased deposits from non-bank bond investors as compensation for selling their bonds to the Fed. These new deposits, as we have also shown, are more likely to be large and uninsured. Because they are generated by Fed actions rather than market forces, interest-rate sensitivity may also be lower for them when unconventional policies are operating. We test these effects next.

Table 3 separates the effect of the extended zero lower bound from that of QE deposit exposure on the sensitivity of deposit flows to the deposit rate gap. To do so, we report estimates of the following specification at the bank-quarter level:

$$\begin{aligned}\Delta \text{Dep}^{k}_{b,t} / \text{Assets}_{b,t-1} = {} & \beta_1 \, \text{Rate Gap}_{b,t} + \beta_2 \, \text{QE Exposure}_{b,t-1} + \beta_3 \, \text{Rate Gap}_{b,t} \times \text{ZLB}_t \\ & + \beta_4 \, \text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1} \\ & + \beta_5 \, \text{QE Exposure}_{b,t-1} \times \text{ZLB}_t \\ & + \beta_6 \, \text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1} \times \text{ZLB}_t \\ & + \alpha^{r}_{b} + \gamma_t + \epsilon_{b,t}\end{aligned} \tag{3}$$

where $\Delta \text{Dep}^{k}_{b,t}$ denotes the quarterly change in deposits of category k for bank b, with k

indexing total, uninsured, and insured deposits across Panels A, B, and C, respectively, each scaled by lagged total assets.

Following Begenau and Stafford (2022), Rate Gap$_{b,t}$ is the difference between bank b's annualized deposit rate (calculated from the Call Reports as interest expense on deposits divided by average deposits) and the deposit-share-weighted rate of competing banks in its operating counties. These deposit weights are predetermined using the most recent June Summary of Deposits — specifically, 2019 June weights for 2020Q1–Q2 and 2020 June weights for 2020Q3–Q4. Under uniform rate setting, Rate Gap captures variation in a bank's relative deposit rate that is plausibly exogenous to its own deposit flows.[17] ZLB$_t$ is an indicator variable for zero-lower-bound episodes. At the quarterly frequency, ZLB$_t$ equals one when the quarterly average federal funds rate is at or below 0.25%. At the annual level (corresponding to the Summary of Deposits reporting year), ZLB$_t$ equals one for years in which at least two quarters satisfy this criterion. QE Exposure$_{b,t-1}$ is a standardized (mean zero, variance 1) twelve-quarter lagged moving average of $Z_{b,t}$, which we have shown affects deposit inflows.[18]

To account for unobserved heterogeneity, we include a separate bank fixed effect (α_b^r) for each regime – ZLB v. non-ZLB – and quarter fixed effects (γ_t); the main effect of ZLB$_t$ is absorbed by the quarter fixed effects. We cluster standard errors two-way at the bank and quarter levels. The bank$\times$regime effects capture the idea that different banks may respond differently to unconventional monetary policy, or they may have different underlying clientele who respond differently.[19]

The coefficient of interest is β_1, which measures the baseline sensitivity of deposit flows to Rate Gap, together with the interaction terms that allow this sensitivity to shift

[17] To validate this claim, we have reconstructed Rate Gap by including only those competing markets where each bank raises less than half of its total deposits — these smaller markets would have the least impact on the bank's deposit choice. These results, reported in Appendix Table A.3, are similar to those reported in Table 3.

[18] We leave out the other two measures of QE exposure (based on reserves and MBS), as neither predicts deposit growth.

[19] Unlike the analysis from Tables 1 and 2, we use a backward-looking moving average of QE exposure to account for the idea that the composition of the stock of a bank's deposits at a point in time will have been affected by its QE exposure over many prior quarters. As such, there is limited within-bank time variation in this smoothed version of QE exposure.

with the monetary regime. The interaction with ZLB_t (β_3) allows the rate sensitivity of deposit flows to differ during periods when the policy rate is at its floor, when rate spreads across banks compress toward zero. The interaction with QE $\text{Exposure}_{b,t-1}$ (β_4) allows the sensitivity to depend on how much of a bank's deposit base was accumulated through QE, and the triple interaction (β_6) allows the QE-exposure effect on rate sensitivity itself to differ between ZLB and non-ZLB periods. Column (1) reports the baseline with the Rate Gap alone; Column (2) adds the ZLB and QE-Exposure interactions; and Column (3) adds the triple interaction. Decomposing the dependent variable by insurance status across the three panels distinguishes the response of the riskier, more run-prone uninsured funding base (Panel B) from that of stable insured funding (Panel C), while Panel A reports the combined total. We also introduce, as a robustness test, an additional set of interaction terms using an indicator set to one during the two quarters affected by the SVB run (Q1 and Q2 of 2023).

During the period without unconventional policy — a period which combines quarters before the GFC, as well as the two interest rate cycles which followed — the results suggest strong interest rate sensitivity of deposits, both for total deposits as well as in the decomposition into insured and uninsured. For example, a standard deviation increase in Rate Gap ($= 0.006$) leads to an increase in deposits of about 0.4% of assets (see Table 3, first row of columns 1 and 2). In contrast, this interest sensitivity collapses for banks with high QE exposure during the ZLB period. Exposure to QE lowers deposit rate sensitivity, but only during the ZLB; we find a positive effect of QE exposure on interest rate sensitivity during other periods (column 3).

The fourth column of Table 3 adds SVB interaction effects. These have little consequence for the impact of either ZLB or QE Exposure on rate sensitivity. We do find, however, that rate sensitivity did increase significantly during these two quarters. This suggests that at least some otherwise "sleepy" depositors woke up during the run and the publicity surrounding it. As a result, deposits tended to flow differentially toward high-rate banks.[20]

[20]This result supports Maingi (2026), who argues that reallocation of deposits across banks in the wake

Panels B and C report similar regressions separately for uninsured and insured deposits. Magnitudes are greater for insured deposits than for uninsured, but the basic patterns are the same. For both types, rate sensitivity falls sharply during the ZLB when banks are highly exposed to QE. One reason for the magnitude difference may be that Rate Gap is only measurable at the level of total deposits, and insured deposits constitute more than 60% of total bank deposits. One stark difference appears in column 4, which shows that during the quarters after the SVB run interest sensitivity fell sharply for uninsured deposits, while it rose sharply for insured deposits. This contrast likely reflects a rational reaction of uninsured depositors, who pulled funds in response to *solvency* concerns rather than based on interest rates. Most of these funds flowed out of small and large banks and into the top-4 banks (because they were perceived as too big to fail) or into insured accounts (because the deposits would be protected by the FDIC). Panel C shows that the flows into insured accounts did so most at banks paying high rates.[21]

[Insert Table 3 Here]

Table 4 extends this analysis by estimating the same specification separately for small and large banks, classified by total assets below or above $10 billion as of year-end 2008. Splitting the sample by size allows the sensitivity of deposit flows to the Rate Gap — and the way that sensitivity shifts with the ZLB and with QE Exposure — to differ across the size distribution, since large and small banks face different competitive environments, deposit franchises, and degrees of exposure to Federal Reserve balance-sheet policy. Panel A reports the small-bank subsample and Panel B the large-bank subsample. The results are statistically strong for small banks but not large, probably due to relatively low power given the granularity of the fixed effects.

of SVB was efficiency enhancing because more capital flowed toward higher return banks (i.e., those paying higher deposit rates and with better lending opportunities).

[21] Note that all of the Top-4 banks (JP Morgan Chase, Citigroup, Bank of America and Wells Fargo) are primary dealers and thus not included in our sample.

[Insert Table 4 Here]

Table 5 re-estimates the rate sensitivity model, decomposing the results of Table 3 to the bank-county-year level. This more granular panel allows us to absorb local demand and market conditions much more flexibly. We estimate the following specification:

$$\begin{aligned}\Delta\text{Dep}_{b,c,t}/\text{Assets}_{b,t-1} = {} & \beta_1 \text{ Rate Gap}_{b,c,t} + \beta_2 \text{ QE Exposure}_{b,c,t-1} + \beta_3 \text{ Rate Gap}_{b,c,t} \times \text{ZLB}_t \\ & + \beta_4 \text{ Rate Gap}_{b,c,t} \times \text{QE Exposure}_{b,c,t-1} \\ & + \beta_5 \text{ QE Exposure}_{b,c,t-1} \times \text{ZLB}_t \\ & + \beta_6 \text{ Rate Gap}_{b,c,t} \times \text{QE Exposure}_{b,c,t-1} \times \text{ZLB}_t \\ & + \theta_{b,t} + \lambda_{c,t} + \eta_{b,c} + \epsilon_{b,c,t}\end{aligned} \tag{4}$$

where $\Delta\text{Dep}_{b,c,t}$ is the annual change in deposits held by bank b in county c from SOD year $t-1$ to t, scaled by lagged total bank assets. Only total deposits are reported at the branch level each June on the FDIC's Summary of Deposits, so we can not decompose into insured and uninsured.

$\text{Rate Gap}_{b,c,t}$ equals the disaggregated difference between the bank's own annualized deposit rate and the deposit-share-weighted rate of competing banks in the same county; these are the building blocks used to construct the bank-level measure in Tables 3 and 4.[22] ZLB_t is an indicator equal to one when the Federal Funds rate is at its effective floor during the Summary of Deposits year, and $\text{QE Exposure}_{b,c,t-1}$ is a standardized three-year lagged moving average of QE-driven deposit inflows scaled by lagged assets, following Yang (2026). We include bank×year fixed effects ($\theta_{b,t}$), county×year fixed effects ($\lambda_{c,t}$), and bank×county fixed effects ($\eta_{b,c}$), and we two-way cluster standard errors at the bank and county levels.[23]

The advantage of the bank-county specification is the county×year fixed effects,

[22] See Appendix Table A.1 for analytical definitions of both $\text{Rate Gap}_{b,t}$ and $\text{Rate Gap}_{b,c,t}$.

[23] Our model differs from Begenau and Stafford (2022) in two ways. First, we introduce the two interactions representing unconventional policy. Second, we aggregate the deposit growth to the county level rather than using branch-level growth because our QE metric varies at that level.

which absorb everything common to a local deposit market in a given year — local income and credit-demand shocks, regional flight-to-safety flows, and any other market-wide conditions — so that β_1 is identified from differences in deposit growth between banks operating in the *same* county in the *same* year but facing different Rate Gaps. The bank×year fixed effects further absorb any bank-level shock common across a bank's markets in a year, and the bank×county fixed effects absorb the time-invariant component of each bank-county relationship. The bank-time effect is particularly crucial because it strips out variation in demand for deposits, which varies at bank-time but not at the bank-county-time level as long as banks can move funds between markets to finance investment.[24] As in Table 3, the interaction terms allow the Rate-Gap sensitivity of local deposit flows to shift with the ZLB regime (β_3), with the bank's QE exposure (β_4), and with the two jointly (β_6). Column (1) reports the baseline; Column (2) adds the ZLB and QE-Exposure interactions; and Column (3) adds the triple interaction.

[Insert Table 5 Here]

The results in Table 5 are consistent with, though sharper than, those estimated at bank-quarter level (Table 3). Bank-county deposits grow faster when Rate Gap is higher (column 1), with a magnitude larger than that of Table 3. A one-standard-deviation increase in rates leads to deposit growth of about 0.75% of assets ($= 0.005 \times 1.5$). The decline in interest sensitivity is also larger than in Table 3, with a decline of about 1/2 during the ZLB. As in the bank-time analysis, the effect of QE exposure also dampens interest sensitivity during the ZLB (when QE itself is active), but amplifies it during the other periods. Said differently, banks which have accumulated more deposits when QE has been bringing in deposits (i.e., during the ZLB years) subsequently experience higher rate sensitivity when monetary conditions normalize. (We develop this idea further below.)[25]

[24] For evidence that bank funds move across local markets to finance investment, see Gilje, Loutskina and Strahan (2016) and Drechsler, Savov and Schnabl (2017).

[25] In Table A.4 we replace Rate Gap with its analog using only savings accounts; these results are similar in sign-pattern to those reported in Table 5, but with somewhat smaller coefficients.

Column (4) of Table 5 introduces the SVB interactions, which in this case are based on indicators set to one during 2023. Consistent with Table 3, the core results change little. Also consistent with Table 4, the SVB shock sharply increases interest rate sensitivity.

Table 6 estimates the same bank-county specification separately for small and large banks, classified by total assets below or above $10 billion as of year-end 2008. As at the bank-quarter level, splitting by size allows the sensitivity of local deposit flows to the rate gap — and the way it shifts with the ZLB and with QE Exposure — to differ across the size distribution, since large and small banks differ in their local market power, deposit franchises, and exposure to Federal Reserve balance-sheet policy. Panel A reports the small-bank subsample and Panel B the large-bank subsample. The estimated rate-sensitivity is greater in the small-bank sample, although sign patterns are similar for both large and small banks. In the latter sample, we find a large negative interaction between Rate Gap and the ZLB indicator; however, the sum of the direct and interaction terms is not statistically significantly different from zero ($= 0.98 - 2.06$; p-value for F-stat $= 0.274$).

[Insert Table 6 Here]

4.3 Deposit Rate Sensitivity when Monetary Conditions Normalize

Tables 3 through 6 show that deposit-flow sensitivity to the Rate Gap falls during the ZLB era when QE Exposure is high. In the best-identified models which fully account for bank-time deposit demand (Tables 5 and 6), the decline in rate sensitivity happens widely across banks. These patterns suggest that the incremental deposits flowing into banks are being distorted by unconventional policy. The ZLB, by creating compression in rates both across banks and between banks and money markets (recall Figures 4 and 6), may draw in funds from normally rate-sensitive customers who are putting their liquidity into banks (rather than money markets) for the increased safety and liquidity. The same argument goes through for deposits pulled into banks by QE. Once rates move off zero, these deposits start to discriminate *more* than other deposits

based on interest rates. By altering the composition of deposits, both dimensions of unconventional policy potentially affect how deposit supply responds once rates move off the zero lower bound.

To test these ideas, we need to capture cross-bank variation in exposure to both QE and to the ZLB. Unlike QE (where we exploited both county and bank-county variation), our prior models captured the ZLB with a simple time-series indicator variable. Since some banks raised more deposits than others during the ZLB quarters, we now exploit this cross-bank heterogeneity in exposure. We measure this variation using each bank's deposit growth accumulated during the ZLB quarters, scaled by assets. To do so, we build a bank-time-varying *ZLB Exposure*, defined as the backward-looking, 12-quarter moving average of the bank's quarterly deposit change scaled by its total assets at the start of the ZLB episode, counting only those quarters that occurred during the ZLB regime (non-ZLB quarters enter as zero). As the moving-average window rolls forward and ZLB quarters are progressively replaced by non-ZLB quarters, the measure tends toward zero; because it averages signed deposit changes, however, it need not decline monotonically. *QE Exposure* is constructed analogously: it is the 12-quarter moving average of bank b's QE-driven deposit inflow, scaled by lagged assets. To ease the comparisons, we standardize both measures to have zero mean and unit variance.

We estimate the following specification, now using only the data during the two rate cycles which followed each ZLB episode:

$$\begin{aligned}\Delta \text{Dep}^{k}_{b,t} / \text{Assets}_{b,t-1} = \beta_1 \, \text{Rate Gap}_{b,t} + \beta_2 \, \text{ZLB Exposure}_{b,t-1} + \beta_3 \, \text{QE Exposure}_{b,t-1} \\ + \beta_4 \, \text{Rate Gap}_{b,t} \times \text{ZLB Exposure}_{b,t-1} \\ + \beta_5 \, \text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1} \\ + \alpha^{r}_{b} + \gamma_t + \epsilon_{b,t}\end{aligned} \tag{5}$$

We estimate this model on the non-ZLB quarters of each episode, so that β_1 captures the rate-gap sensitivity of deposit flows once the policy rate has lifted off its floor, β_4 captures how that sensitivity differs across banks that accumulated more or less

deposits during the preceding ZLB episode, and β_5 measures the analogous differential with respect to accumulated QE Exposure. While some of the deposit inflows driving ZLB Exposure came from QE bond purchases, by controlling for QE Exposure we can interpret β_4 as capturing the marginal effect of the ZLB.[26] The specification includes a separate bank fixed effect for each of the two cycles (α_b^r) and quarter fixed effects (γ_t). The dependent variables $\Delta \text{Dep}_{b,t}^k$ are based on total deposits or uninsured deposits. We cluster standard errors two-way by bank and state-quarter.

A causal reading of this model is complicated by the endogeneity of ZLB Exposure. Banks choose the amount of deposits raised in response to lending and other investment opportunities.[27] If these (unmeasured) investment opportunities are persistent, they could affect our estimates of rate sensitivity.

We address this endogeneity concern by replacing each bank's ZLB Exposure with predetermined, plausibly exogenous variation in its exposure to the ZLB episode. Our instrument is a leave-out shift-share measure (Bartik, 1991). As noted above, the ZLB led to compositional shift away from time deposits and toward non-interest bearing deposits (recall Figure 3). As such, banks which entered a ZLB era with high levels of either type would be highly exposed to the policy. To build the measure ($ZLB\ Exposure_{b,t}^{Bartik}$), we compute the national growth rate of each of the three deposit types (time deposits, savings deposits and non-interest bearing deposits) in every ZLB quarter (excluding the bank's own deposits) and then combine these using weights equal to the bank's holdings of each deposit type divided by its total assets at the start of the episode. The resulting predicted inflow is therefore scaled by assets, as is *ZLB Exposure*, and the weights sum to the bank's pre-episode ratio of these deposits to assets rather than to one. Both the weights and the accumulation window are specific to each episode: in the

[26] In fact, as shown in Table A.5, QE exposure is not correlated with ZLB exposure across banks in both episodes. Both dimensions of unconventional policy are strongly (and independently) correlated with deposit composition. As shown in Table A.6, both ZLB exposure and QE exposure positively predict both the share of deposits in non-interest-bearing accounts as well as the share of uninsured deposits.

[27] In contrast, QE Exposure depends only on actions taken by the Fed in combination with the markets in which each primary dealer operates. Yang (2026) considers the possibility that endogeneity in this measure stems from the location of investors willing to sell bonds to the Fed. However, the results there provide no evidence that this channel is important.

first, we use holdings as of 2008Q2 and accumulate over the post-GFC ZLB quarters; in the second, we use holdings as of 2019Q4 and accumulate over the pandemic-era ZLB quarters. Hence, the instrument for each rate cycle is predetermined relative to that episode's shock. The identifying assumption is that a bank's exposure to the *aggregate* ZLB-era deposit shift is uncorrelated with the bank's own subsequent rate-setting except through its effect on accumulated ZLB deposits. We interact this Bartik instrument with the Rate Gap to replace the endogenous Rate-Gap$\times$ZLB-Exposure interaction (i.e., we report the reduced form).

Panel A of Table 7 reports two specifications for total deposits and two for uninsured deposits. Odd columns report OLS and even ones are reduced form, meaning the Bartik instrument replaces the bank's own ZLB Exposure. Both ZLB Exposure and QE Exposure are standardized across the full, pooled non-ZLB sample. In all four cases, QE Exposure raises interest rate sensitivity. For total deposit growth, banks with QE exposure one standard deviation higher than average face rate sensitivity which is about 50 percent larger than average (0.24/0.44). That is, banks more exposed to QE needed to raise their deposit rates more to retain them, once policy normalized. Exposure to the ZLB also raised interest-rate sensitivity, with consistent effects in the reduced form, although smaller in magnitude than QE exposure. For the OLS model, we do not find a significant effect of ZLB Exposure on uninsured deposit growth interest sensitivity.[28]

[Insert Table 7 Here]

Panel B of Table 7 estimates similar models, now at the bank-county-year level rather than the bank-quarter level; the finer geography lets us absorb local deposit-market conditions much more flexibly and more fully. Moreover, we can remove concerns about deposit demand correlating with the ZLB Exposure by estimation within bank-year. As

[28]These results are robust to only using the "small markets", which are shown in Appendix Table A.7.

such, we report OLS estimates of the following model:

$$\begin{aligned}\Delta \text{Dep}_{b,c,t}/\text{Assets}_{b,t-1} = \beta_1 \text{ Rate Gap}_{b,c,t} + \beta_2 \text{ ZLB Exposure}_{b,c,t-1} + \beta_3 \text{ QE Exposure}_{b,c,t-1} \\ + \beta_4 \text{ Rate Gap}_{b,c,t} \times \text{ZLB Exposure}_{b,c,t-1} \\ + \beta_5 \text{ Rate Gap}_{b,c,t} \times \text{QE Exposure}_{b,c,t-1} \\ + \theta_{b,t} + \lambda_{c,t} + \epsilon_{b,c,t}\end{aligned} \tag{6}$$

where $\Delta \text{Dep}_{b,c,t}$ is the annual change in bank b's deposits in county c, scaled by lagged bank assets; the FDIC Summary of Deposits reports only total deposits at this level (meaning total deposits at annual frequency), so the bank-county analysis has a single dependent variable and can only use annual frequency. QE Exposure and ZLB Exposure are computed analogously to these, measured at the bank-county-year level, although now we look back three years rather than 12 quarters. While the Bartik measure of ZLB exposure is not available at this level of aggregation (because we can not measure deposit changes by type at the bank-county level), we can absorb variation in demand for deposits by absorbing granular fixed effects at both the bank-year and county (or county-year) levels.

Equation (6) is again estimated including only data from the two rate cycles following each ZLB episode, so that β_1 is the local rate-gap sensitivity of deposit flows once the policy rate has lifted off zero, β_4 captures how that sensitivity differs across bank-counties that accumulated more or less deposits during the preceding ZLB spell, and β_5 the analogous differential with respect to accumulated QE exposure. We cluster standard errors two-way by bank and county. The findings are consistent with those at the bank-quarter level, with exposure to both dimensions of unconventional policy leading to higher rate sensivitity.[29] Overall, both dimensions of unconventional monetary policy led to flows of deposits into banks from normally rate-sensitive customers. These deposits were stable as long as rates across banks and between banks and the money market remained compressed. When policy normalized with an end to QE and

[29] Appendix Table A.8 estimates these models after dropping the SVB crisis period. These demonstrate that the effects on interest sensitivity of exposure to both the ZLB and QE are robust.

an increase in the Fed's policy rate, banks needed to increase deposit rates aggressively in order to keep those deposits.

Our last test considers this issue: how did bank deposit pricing — the deposit beta — respond to exposure to unconventional monetary policy? This construct, developed in Drechsler, Savov and Schnabl (2017), measures how much a bank increases its deposit rate per unit of increase in money market rates. They build the deposit beta, bank by bank, from the time series during the period between 1994 and 2013, and find an average of 0.54.[30] We estimate betas using a bank-quarter panel regression, with the change in the deposit spread (= Fed Funds rate − deposit rate) regressed on three lags of changes in the Federal Funds rate interacted with both QE Exposure and ZLB Exposure, as well as a set of control variables that may affect the deposit beta — a bank's lagged log total assets, its holdings of non-interest-bearing, savings, and time deposits at the start of the ZLB episode (each scaled by total assets at that date), and, importantly, the bank's pre-episode HHI, each interacted with the change in the Fed Funds rate — as follows:

$$\begin{aligned}\Delta\text{Dep Spread}_{b,t} = {} & \beta_1^e \,\text{ZLB Exposure}_{b,t-1} + \beta_2 \,\text{QE Exposure}_{b,t-1} \\ & + \sum_{h=0}^{3} \beta_{3h} \,\text{ZLB Exposure}_{b,t-1} \times \Delta\text{FFR}_{t-h} \\ & + \sum_{h=0}^{3} \beta_{4h} \,\text{QE Exposure}_{b,t-1} \times \Delta\text{FFR}_{t-h} \\ & + \sum_{h=0}^{3} X'_{b,t-1}\,\Gamma_h \times \Delta\text{FFR}_{t-h} + \alpha_b + \gamma_t + \epsilon_{b,t},\end{aligned} \tag{7}$$

where $\Delta\text{Dep Spread}_{b,t}$ is the quarterly change in bank b's deposit spread.

As in Equation (6), we estimate the model using only data from the two rate cycles that followed unconventional policy (2016Q1–2020Q1 and 2022Q2–2025Q4). We also report the change in the overall deposit spread, as well as one built from savings deposits and from time deposits. Notably, we report the effects of policy net of the bank's own market power over deposits, a key determinant emphasized by Drechsler,

[30] Narayanan, Ratnadiwakara and Strahan (2025) estimate betas from the cross-section and find somewhat lower values during the more recent rate cycle (closer to 0.3). Kang-Landsberg, Luck and Plosser (2023) also report a beta of about 0.3 during the rate cycle starting in 2016.

Savov and Schnabl (2017).

[Insert Table 8 Here]

Table 8 reports the sum of the coefficients interacting the change in the Fed Funds rate and its three lags. Since the dependent variable equals the change in the deposit spread, a negative coefficient implies that an increase in ZLB/QE Exposure leads to *higher* deposit betas, and this is precisely what we find. Both policies raise beta, although the effect of ZLB Exposure is both larger in magnitude and statistical significance compared to QE exposure; the effects are consistent in both the OLS models (Panel A) compared to the reduced form (Panel B). From Panel A, column 1, a one standard deviation increase in ZLB Exposure would increase the deposit beta by 0.019, compared to its average of about 0.3 during this period (Narayanan, Ratnadiwakara and Strahan (2025)). These patterns align with the flow results: banks more exposed to unconventional policy experience greater rate sensitivity of deposit supply; they react by raising deposit rates more with increases in money market rates than other banks to retain some of those deposits.

5 Conclusion

Both dimensions of unconventional monetary policy, we find, came with the unexpected consequence of making deposits both costlier and riskier to retain once policy normalized. When the Fed decided to lower the short-term policy rate to zero and committed to leave it there, it did so following a financial crisis sufficiently reminiscent of the 1930s to make a catastrophic deflationary spiral plausible. Adding additional monetary stimulus through its large-scale purchase of long-term bonds also seemed well motivated at the time. But these policies led to very large expansions of bank deposits from new sources; deposits, for example, grew by 42% between 2010 and 2015 (the first instance of the ZLB), while nominal GDP grew just 26%. These inflows tilted the composition of bank funding toward more rate-sensitive sources, such as corporate

Treasuries which might otherwise park liquidity in the money markets. We exploit heterogeneity in bank exposure to both QE and the ZLB to show that this compositional change meant that banks had to raise deposits rates more to keep those deposits once monetary conditions normalized. If future circumstances demand unconventional responses, policy makers should take into account their effects on bank funding stability.

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Table 1: Alternative QE Exposure Measures and Deposit Flows by Category

This table reports estimates of the effect of alternative QE exposure measures on banks' deposits by category. Panels A, B, and C examine the change in total, uninsured, and insured deposits, respectively, each scaled by lagged total assets. The independent variables are a bank-level reserve instrument ($Z^{R1}_{b,t}$) following Acharya et al. (2024), a shift-share instrument capturing the direct asset-swap effect ($Z_{\text{mbs},b,t}$), and the deposit channel instrument ($Z_{b,t}$) following Yang (2026). Within each sample period, the first column excludes the reserve instrument and the second includes it. The reserve instrument is measured in percent; the remaining instruments are in dollars, scaled by the prior quarter's total bank-level assets. The four sample periods are the Full Sample (2009–2025), QE I–III (2009–2014), Post-QE + QT (2015–2019), and Pandemic QE (2020–2021). All variables are winsorized at the 1st and 99th percentiles, and all specifications include bank and quarter fixed effects. Standard errors are two-way clustered by bank and quarter in the Full Sample columns (which span more than 30 quarters) and by bank and state-quarter in the sub-period columns; clustered standard errors are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Total Deposits

	Full Sample		QE I–III		Post QE + QT		Pandemic QE	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent Variable:	$\Delta\text{Dep}_{b,t}/\text{Assets}_{b,t-1}$							
$Z^{R1}_{b,t}$		0.02		-0.07		0.00		-0.04
		(0.21)		(0.17)		(0.18)		(0.29)
$Z_{\text{mbs},b,t}/\text{Assets}_{b,t-1}$	0.15*	0.24*	-0.20**	-0.28**	-0.06	-0.10	0.47**	0.94**
	(0.08)	(0.13)	(0.08)	(0.12)	(0.19)	(0.29)	(0.20)	(0.47)
$Z_{b,t}/\text{Assets}_{b,t-1}$	0.24	0.16	0.78***	0.65***	0.07	0.28	1.65***	1.44***
	(0.16)	(0.11)	(0.11)	(0.11)	(0.17)	(0.25)	(0.21)	(0.47)
N	392,851	114,875	166,247	52,700	114,168	40,323	39,519	7,880
R^2	0.13	0.14	0.11	0.13	0.08	0.09	0.27	0.28
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 1 continued from previous page

Panel B: Uninsured Deposits

	Full Sample		QE I–III		Post QE + QT		Pandemic QE	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent Variable:	ΔUninsured $\text{Dep}_{b,t}/\text{Assets}_{b,t-1}$							
$Z^{R1}_{b,t}$		0.25		0.26*		0.04		0.09
		(0.17)		(0.14)		(0.13)		(0.21)
$Z_{\text{mbs},b,t}/\text{Assets}_{b,t-1}$	0.07	0.13*	-0.06	-0.11	0.05	-0.12	0.23	0.52
	(0.05)	(0.07)	(0.07)	(0.10)	(0.15)	(0.21)	(0.16)	(0.32)
$Z_{b,t}/\text{Assets}_{b,t-1}$	0.26**	0.20**	0.50***	0.45***	0.10	0.32	1.22***	1.34***
	(0.12)	(0.09)	(0.09)	(0.09)	(0.14)	(0.19)	(0.19)	(0.32)
N	392,851	114,875	166,247	52,700	114,168	40,323	39,519	7,880
R^2	0.23	0.26	0.34	0.36	0.00	0.02	0.08	0.16
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Panel C: Insured Deposits

	Full Sample		QE I–III		Post QE + QT		Pandemic QE	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent Variable:	ΔInsured $\text{Dep}_{b,t}/\text{Assets}_{b,t-1}$							
$Z^{R1}_{b,t}$		-0.23*		-0.33***		-0.04		-0.13
		(0.13)		(0.11)		(0.08)		(0.14)
$Z_{\text{mbs},b,t}/\text{Assets}_{b,t-1}$	0.07	0.10	-0.14***	-0.18**	-0.11	0.02	0.25**	0.42
	(0.05)	(0.08)	(0.05)	(0.08)	(0.11)	(0.18)	(0.11)	(0.26)
$Z_{b,t}/\text{Assets}_{b,t-1}$	-0.02	-0.04	0.28***	0.20**	-0.03	-0.04	0.42***	0.10
	(0.07)	(0.07)	(0.08)	(0.08)	(0.10)	(0.12)	(0.08)	(0.23)
N	392,851	114,875	166,247	52,700	114,168	40,323	39,519	7,880
R^2	0.36	0.36	0.49	0.49	0.08	0.10	0.27	0.23
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 2: Deposit Channel Instrument Relevance for Bank-County Deposits

This table reports first-stage regression results at the bank-county-year level. The dependent variable is the annual change in bank-county deposits, and the independent variable is the deposit channel instrument ($Z_{b,c,t}$), following Yang (2026). Both the dependent variable and the deposit channel instrument are scaled by the previous year's total bank-level assets and winsorized at the 1st and 99th percentiles. Column (1) covers the full-sample period (2009–2025). Column (2) restricts the sample to small banks, defined as those with total assets below $10 billion as of year-end 2008. Columns (3) and (4) report results for the QE I–III (2009–2014) and Pandemic QE (2020–2021) sub-periods, respectively. All specifications include bank×year and county×year fixed effects; Columns (1) and (2) further include bank×county fixed effects. Robust standard errors, two-way clustered by bank and county, are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)	(4)
Dependent Variable:	$\Delta\text{Dep}_{b,c,t}/\text{Assets}_{b,t-1}$			
$Z_{b,c,t}/\text{Assets}_{b,t-1}$	0.89***	0.95***	0.25***	1.92***
	(0.07)	(0.08)	(0.07)	(0.08)
N	279,069	207,378	91,873	34,736
R^2	0.39	0.33	0.32	0.42
F-stat	161.15	157.97	13.82	531.08
Bank×Year FE	Yes	Yes	Yes	Yes
County×Year FE	Yes	Yes	Yes	Yes
Bank×County FE	Yes	Yes	No	No
Bank Size	All	Small	All	All
Sample Period	Full Sample	Full Sample	QE I–III	Pandemic QE

Table 3: Separating the Effects of the ZLB and QE on Deposit-Flow Sensitivity to the Rate Gap (Bank-Quarter)

This table separates the effect of the zero lower bound (ZLB) from that of QE exposure on the sensitivity of deposit flows to the deposit rate gap, at the bank-quarter level over 2001Q1–2025Q4. Panels A, B, and C split the dependent variable into total, uninsured, and insured deposits, respectively. The Rate Gap, following Begenau and Stafford (2022), is the difference between the bank's own annualized deposit rate and the deposit-share-weighted rate of competing banks in the bank's counties, using predetermined deposit weights. ZLB_t equals one when the federal funds rate is at its effective floor. SVB_t equals one in 2023Q1–2023Q2, capturing the deposit disruption around the failure of Silicon Valley Bank. $\text{QE Exposure}_{b,t-1}$ is a standardized twelve-quarter lagged moving average of QE-driven deposit inflows scaled by lagged assets, following Yang (2026). Column (1) is the baseline; Column (2) adds the ZLB and QE-Exposure interactions; Column (3) adds the triple interaction; Column (4) adds the SVB interactions and the triple interaction. All specifications include bank×regime and quarter fixed effects, where the regime denotes the ZLB versus non-ZLB monetary environment. The main effects of ZLB_t and SVB_t are absorbed. Robust standard errors, clustered two-way by bank and quarter, are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Total Deposits

	(1)	(2)	(3)	(4)
Dependent Variable:	$\Delta\text{Dep}_{b,t}/\text{Assets}_{b,t-1}$			
$\text{Rate Gap}_{b,t}$	0.59***	0.58***	0.60***	0.58***
	(0.08)	(0.08)	(0.07)	(0.08)
$\text{QE Exposure}_{b,t-1}$		-0.001**	-0.001**	-0.001**
		(0.000)	(0.000)	(0.000)
$\text{Rate Gap}_{b,t} \times \text{ZLB}_t$		0.11	0.03	0.06
		(0.31)	(0.28)	(0.28)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1}$		-0.00	0.13**	0.10
		(0.07)	(0.06)	(0.08)
$\text{QE Exposure}_{b,t-1} \times \text{ZLB}_t$		0.003***	0.004***	0.004***
		(0.001)	(0.001)	(0.001)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1} \times \text{ZLB}_t$			-0.57***	-0.54***
			(0.14)	(0.15)
$\text{Rate Gap}_{b,t} \times \text{SVB}_t$				0.63***
				(0.13)
$\text{QE Exposure}_{b,t-1} \times \text{SVB}_t$				-0.000
				(0.001)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1} \times \text{SVB}_t$				-0.05
				(0.06)
N	629,422	618,914	618,914	618,914
Bank×Regime FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes

Table 3 continued from previous page

Panel B: Uninsured Deposits

	(1)	(2)	(3)	(4)
Dependent Variable:	ΔUninsured $\text{Dep}_{b,t}/\text{Assets}_{b,t-1}$			
$\text{Rate Gap}_{b,t}$	0.17***	0.16***	0.17***	0.18***
	(0.06)	(0.04)	(0.04)	(0.04)
$\text{QE Exposure}_{b,t-1}$		-0.001***	-0.001***	-0.001***
		(0.000)	(0.000)	(0.000)
$\text{Rate Gap}_{b,t} \times \text{ZLB}_t$		0.04	-0.00	-0.00
		(0.27)	(0.26)	(0.26)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1}$		0.00	0.07**	0.06*
		(0.04)	(0.03)	(0.04)
$\text{QE Exposure}_{b,t-1} \times \text{ZLB}_t$		0.002***	0.002***	0.002***
		(0.001)	(0.001)	(0.001)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1} \times \text{ZLB}_t$			-0.29***	-0.28***
			(0.10)	(0.10)
$\text{Rate Gap}_{b,t} \times \text{SVB}_t$				-0.12*
				(0.06)
$\text{QE Exposure}_{b,t-1} \times \text{SVB}_t$				-0.001
				(0.001)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1} \times \text{SVB}_t$				0.03
				(0.03)
N	629,422	618,914	618,914	618,914
Bank×Regime FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes

Panel C: Insured Deposits

	(1)	(2)	(3)	(4)
Dependent Variable:	ΔInsured $\text{Dep}_{b,t}/\text{Assets}_{b,t-1}$			
$\text{Rate Gap}_{b,t}$	0.42***	0.41***	0.43***	0.40***
	(0.05)	(0.04)	(0.04)	(0.04)
$\text{QE Exposure}_{b,t-1}$		-0.000	-0.000	-0.000
		(0.000)	(0.000)	(0.000)
$\text{Rate Gap}_{b,t} \times \text{ZLB}_t$		0.07	0.03	0.06
		(0.19)	(0.18)	(0.18)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1}$		-0.01	0.06	0.03
		(0.04)	(0.04)	(0.04)
$\text{QE Exposure}_{b,t-1} \times \text{ZLB}_t$		0.001***	0.002***	0.002***
		(0.000)	(0.000)	(0.000)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1} \times \text{ZLB}_t$			-0.28***	-0.26***
			(0.08)	(0.08)
$\text{Rate Gap}_{b,t} \times \text{SVB}_t$				0.75***
				(0.15)
$\text{QE Exposure}_{b,t-1} \times \text{SVB}_t$				0.001
				(0.000)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1} \times \text{SVB}_t$				-0.08*
				(0.04)
N	629,422	618,914	618,914	618,914
Bank×Regime FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes

Table 4: Separating the Effects of the ZLB and QE on Deposit-Flow Sensitivity, by Bank Size (Bank-Quarter)

This table reports the specification of Table 3 for total deposits, split by bank size, at the bank-quarter level over 2001Q1–2025Q4. Panel A restricts the sample to small banks, defined as those with total assets below $10 billion as of year-end 2008, and Panel B to large banks, those at or above $10 billion. Variable definitions follow Table 3. SVB_t equals one in 2023Q1–2023Q2, capturing the deposit disruption around the failure of Silicon Valley Bank. Column (1) is the baseline; Column (2) adds the ZLB and QE-Exposure interactions; Column (3) adds the triple interaction; Column (4) adds the SVB interactions and the triple interaction. All specifications include bank×regime and quarter fixed effects, where the regime denotes the ZLB versus non-ZLB monetary environment. The main effects of ZLB_t and SVB_t are absorbed. Robust standard errors, clustered two-way by bank and quarter, are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Small Banks

	(1)	(2)	(3)	(4)
Dependent Variable:	$\Delta\text{Dep}_{b,t} / \text{Assets}_{b,t-1}$			
$\text{Rate Gap}_{b,t}$	0.59***	0.56***	0.58***	0.56***
	(0.09)	(0.08)	(0.08)	(0.08)
$\text{QE Exposure}_{b,t-1}$		-0.001***	-0.001***	-0.001***
		(0.000)	(0.000)	(0.000)
$\text{Rate Gap}_{b,t} \times \text{ZLB}_t$		0.19	0.10	0.12
		(0.30)	(0.27)	(0.27)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1}$		-0.03	0.10*	0.07
		(0.07)	(0.06)	(0.08)
$\text{QE Exposure}_{b,t-1} \times \text{ZLB}_t$		0.003***	0.004***	0.004***
		(0.001)	(0.001)	(0.001)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1} \times \text{ZLB}_t$			-0.55***	-0.52***
			(0.14)	(0.15)
$\text{Rate Gap}_{b,t} \times \text{SVB}_t$				0.67***
				(0.18)
$\text{QE Exposure}_{b,t-1} \times \text{SVB}_t$				-0.000
				(0.001)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1} \times \text{SVB}_t$				-0.06
				(0.07)
N	558,930	551,648	551,648	551,648
Bank×Regime FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes

Table 4 continued from previous page

Panel B: Large Banks

	(1)	(2)	(3)	(4)
Dependent Variable:	$\Delta \text{Dep}_{b,t} / \text{Assets}_{b,t-1}$			
$\text{Rate Gap}_{b,t}$	0.20	0.30	0.29	0.25
	(0.31)	(0.32)	(0.31)	(0.31)
$\text{QE Exposure}_{b,t-1}$		0.001	0.001	0.001
		(0.002)	(0.001)	(0.001)
$\text{Rate Gap}_{b,t} \times \text{ZLB}_t$		-0.18	-0.16	-0.12
		(0.73)	(0.74)	(0.74)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1}$		0.26	0.23	0.11
		(0.22)	(0.16)	(0.12)
$\text{QE Exposure}_{b,t-1} \times \text{ZLB}_t$		0.002	0.002	0.002
		(0.002)	(0.002)	(0.003)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1} \times \text{ZLB}_t$			0.11	0.22
			(0.67)	(0.65)
$\text{Rate Gap}_{b,t} \times \text{SVB}_t$				-0.02
				(0.41)
$\text{QE Exposure}_{b,t-1} \times \text{SVB}_t$				0.002
				(0.001)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1} \times \text{SVB}_t$				0.30
				(0.39)
N	6,482	6,395	6,395	6,395
Bank×Regime FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes

Table 5: Separating the Effects of the ZLB and QE on Local Deposit-Flow Sensitivity to the Rate Gap (Bank-County-Year)

This table separates the effect of the zero lower bound (ZLB) from that of QE exposure on the sensitivity of local deposit flows to the Rate Gap, at the bank-county-year level over the 2001–2025 sample period. The dependent variable is the annual change in bank-county deposits, from the FDIC Summary of Deposits, scaled by lagged bank assets. The Rate Gap, following Begenau and Stafford (2022), is the difference between the bank's own annualized deposit rate and the deposit-share-weighted rate of competing banks in the county, using predetermined deposit weights. ZLB_t equals one when the Federal Funds rate is at its effective floor during the SOD year. SVB_t equals one in the 2023 SOD year, capturing the deposit disruption around the failure of Silicon Valley Bank. $\text{QE Exposure}_{b,c,t-1}$ is a standardized three-year lagged moving average of QE-driven deposit inflows scaled by lagged assets, following Yang (2026). Column (1) is the baseline; Column (2) adds the ZLB and QE-Exposure interactions; Column (3) adds the triple interaction; Column (4) adds the SVB interactions and the triple interaction. All specifications include bank×year, county×year, and bank×county fixed effects; the main effects of ZLB_t and SVB_t are absorbed. Robust standard errors, clustered two-way by bank and county, are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)	(4)
Dependent Variable:	$\Delta\text{Dep}_{b,c,t} / \text{Assets}_{b,t-1}$			
$\text{Rate Gap}_{b,c,t}$	1.52***	1.71***	1.74***	1.49***
	(0.15)	(0.16)	(0.16)	(0.16)
$\text{QE Exposure}_{b,c,t-1}$		-0.003***	-0.003***	-0.002***
		(0.000)	(0.000)	(0.000)
$\text{Rate Gap}_{b,c,t} \times \text{ZLB}_t$		-0.80***	-0.85***	-0.63**
		(0.29)	(0.30)	(0.31)
$\text{Rate Gap}_{b,c,t} \times \text{QE Exposure}_{b,c,t-1}$		0.14***	0.30***	0.13*
		(0.05)	(0.06)	(0.07)
$\text{QE Exposure}_{b,c,t-1} \times \text{ZLB}_t$		0.001***	0.002***	0.001*
		(0.000)	(0.000)	(0.000)
$\text{Rate Gap}_{b,c,t} \times \text{QE Exposure}_{b,c,t-1} \times \text{ZLB}_t$			-0.64***	-0.49***
			(0.10)	(0.12)
$\text{Rate Gap}_{b,c,t} \times \text{SVB}_t$				1.16***
				(0.33)
$\text{QE Exposure}_{b,c,t-1} \times \text{SVB}_t$				-0.003***
				(0.001)
$\text{Rate Gap}_{b,c,t} \times \text{QE Exposure}_{b,c,t-1} \times \text{SVB}_t$				0.23***
				(0.08)
N	366,462	332,205	332,205	332,205
Bank×Year FE	Yes	Yes	Yes	Yes
County×Year FE	Yes	Yes	Yes	Yes
Bank×County FE	Yes	Yes	Yes	Yes

Table 6: Separating the Effects of the ZLB and QE on Local Deposit-Flow Sensitivity, by Bank Size (Bank-County-Year)

This table reports the specification of Table 5, split by bank size, at the bank-county-year level over the 2001–2025 sample period. Panel A restricts the sample to small banks, defined as those with total assets below $10 billion as of year-end 2008, and Panel B to large banks, those at or above $10 billion. Variable definitions and specifications follow Table 5. All specifications include bank×year, county×year, and bank×county fixed effects; the main effects of ZLB_t and SVB_t are absorbed. Robust standard errors, clustered two-way by bank and county, are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Small Banks				
	(1)	(2)	(3)	(4)
Dependent Variable:	$\Delta Dep_{b,c,t} / Assets_{b,t-1}$			
$Rate\ Gap_{b,c,t}$	1.81***	1.94***	1.96***	1.68***
	(0.18)	(0.19)	(0.19)	(0.19)
$QE\ Exposure_{b,c,t-1}$		-0.004***	-0.004***	-0.002***
		(0.000)	(0.000)	(0.001)
$Rate\ Gap_{b,c,t} \times ZLB_t$		-0.73**	-0.78**	-0.53
		(0.34)	(0.34)	(0.35)
$Rate\ Gap_{b,c,t} \times QE\ Exposure_{b,c,t-1}$		0.12**	0.29***	0.10
		(0.06)	(0.07)	(0.09)
$QE\ Exposure_{b,c,t-1} \times ZLB_t$		0.002***	0.002***	0.001
		(0.000)	(0.000)	(0.000)
$Rate\ Gap_{b,c,t} \times QE\ Exposure_{b,c,t-1} \times ZLB_t$			-0.68***	-0.51***
			(0.12)	(0.14)
$Rate\ Gap_{b,c,t} \times SVB_t$				1.23***
				(0.37)
$QE\ Exposure_{b,c,t-1} \times SVB_t$				-0.003***
				(0.001)
$Rate\ Gap_{b,c,t} \times QE\ Exposure_{b,c,t-1} \times SVB_t$				0.25***
				(0.09)
N	266,209	248,917	248,917	248,917
Bank×Year FE	Yes	Yes	Yes	Yes
County×Year FE	Yes	Yes	Yes	Yes
Bank×County FE	Yes	Yes	Yes	Yes

Table 6 continued from previous page

Panel B: Large Banks

	(1)	(2)	(3)	(4)
Dependent Variable:	$\Delta \text{Dep}_{b,c,t} / \text{Assets}_{b,t-1}$			
$\text{Rate Gap}_{b,c,t}$	0.74	0.93	0.98	0.92
	(0.55)	(0.74)	(0.74)	(0.81)
$\text{QE Exposure}_{b,c,t-1}$		-0.003*	-0.004***	-0.005*
		(0.002)	(0.001)	(0.003)
$\text{Rate Gap}_{b,c,t} \times \text{ZLB}_t$		-2.00**	-2.06**	-2.04**
		(0.98)	(0.98)	(1.02)
$\text{Rate Gap}_{b,c,t} \times \text{QE Exposure}_{b,c,t-1}$		-0.01	0.21	0.19
		(0.16)	(0.20)	(0.33)
$\text{QE Exposure}_{b,c,t-1} \times \text{ZLB}_t$		-0.001	0.000	0.001
		(0.002)	(0.002)	(0.002)
$\text{Rate Gap}_{b,c,t} \times \text{QE Exposure}_{b,c,t-1} \times \text{ZLB}_t$			-0.71**	-0.70*
			(0.28)	(0.39)
$\text{Rate Gap}_{b,c,t} \times \text{SVB}_t$				0.30
				(1.67)
$\text{QE Exposure}_{b,c,t-1} \times \text{SVB}_t$				0.003
				(0.004)
$\text{Rate Gap}_{b,c,t} \times \text{QE Exposure}_{b,c,t-1} \times \text{SVB}_t$				0.15
				(0.53)
N	40,937	32,982	32,982	32,982
Bank×Year FE	Yes	Yes	Yes	Yes
County×Year FE	Yes	Yes	Yes	Yes
Bank×County FE	Yes	Yes	Yes	Yes

Table 7: Separating the Effects of the ZLB and QE on Deposit-Flow Sensitivity to the Rate Gap during Non-ZLB Periods

This table reports estimates of deposit-flow sensitivity to the Rate Gap during non-ZLB periods, pooling the post-GFC and post-pandemic episodes. Panel A uses bank-quarter Call Report data (2016Q1–2020Q1 and 2022Q2–2025Q4); the dependent variable is the quarterly change in total (columns 1–2) or uninsured (columns 3–4) deposits, scaled by lagged assets. Panel B uses bank-county-year Summary of Deposits (SOD) data (SOD years 2017–2020 and 2023–2025); the dependent variable is the annual change in bank b's deposits in county c, scaled by the bank's lagged total assets, and Rate $\text{Gap}_{b,c,t}$ is the difference between the bank's deposit rate and the deposit-weighted rate of competing banks in county c. ZLB Exposure is the trailing moving average (twelve quarters in Panel A; three SOD years in Panel B) of the bank's deposit change (in county c in Panel B), counting only ZLB periods (non-ZLB periods contribute zero), scaled by the bank's total assets at the start of the ZLB episode (2008Q2 or 2019Q4 in Panel A; SOD year 2008 or 2019 in Panel B). ZLB $\text{Exposure}^{\text{Bartik}}_{b,t-1}$ is its shift-share instrument: in each ZLB quarter, the bank's holdings of non-interest-bearing, savings, and time deposits at the start of the episode are multiplied by the national growth rate of the corresponding deposit type, computed over all other banks, summed across types, scaled by the bank's total assets at the start of the episode, and averaged over the same twelve-quarter window. QE Exposure is the corresponding trailing moving average of QE-driven inflows, scaled by lagged assets. All exposure measures are standardized over non-ZLB periods and enter both directly and interacted with the Rate Gap. In Panel A, the OLS columns use the endogenous ZLB $\text{Exposure}_{b,t-1}$ and the reduced-form columns replace it with ZLB $\text{Exposure}^{\text{Bartik}}_{b,t-1}$. Panel A includes bank×episode and quarter fixed effects, with standard errors clustered two-way by bank and state-quarter. Panel B includes bank×episode fixed effects together with county×year (column 1) or county (column 2) fixed effects, with standard errors clustered two-way by bank and county. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Bank-Quarter

	Total Deposits		Uninsured Deposits	
	(1) OLS	(2) Reduced Form	(3) OLS	(4) Reduced Form
Dependent Variable:	$\Delta\text{Dep}_{b,t}/\text{Assets}_{b,t-1}$		$\Delta\text{Uninsured Dep}_{b,t}/\text{Assets}_{b,t-1}$	
$\text{Rate Gap}_{b,t}$	0.44***	0.59***	0.30***	0.48***
	(0.07)	(0.08)	(0.05)	(0.06)
$\text{ZLB Exposure}_{b,t-1}$	0.002***		0.002***	
	(0.000)		(0.000)	
$\text{ZLB Exposure}^{\text{Bartik}}_{b,t-1}$		-0.002***		-0.002***
		(0.000)		(0.000)
$\text{QE Exposure}_{b,t-1}$	0.000	0.000*	-0.000***	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)
$\text{Rate Gap}_{b,t} \times \text{ZLB Exposure}_{b,t-1}$	0.11***		-0.04	
	(0.03)		(0.03)	
$\text{Rate Gap}_{b,t} \times \text{ZLB Exposure}^{\text{Bartik}}_{b,t-1}$		0.08**		0.11***
		(0.04)		(0.03)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1}$	0.24***	0.25***	0.14***	0.09***
	(0.03)	(0.03)	(0.02)	(0.02)
N	160,545	148,363	160,543	148,363
Bank×Episode FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes

Table 7 continued from previous page

Panel B: Bank-County-Year

	Total Deposits	
	(1)	(2)
Dependent Variable:	$\Delta \text{Dep}_{b,c,t} / \text{Assets}_{b,t-1}$	
$\text{Rate Gap}_{b,c,t}$	3.12***	2.15***
	(0.16)	(0.13)
$\text{ZLB Exposure}_{b,c,t-1}$	0.000	-0.002***
	(0.000)	(0.000)
$\text{QE Exposure}_{b,c,t-1}$	0.001***	0.001**
	(0.000)	(0.000)
$\text{Rate Gap}_{b,c,t} \times \text{ZLB Exposure}_{b,c,t-1}$	0.21***	0.29***
	(0.04)	(0.04)
$\text{Rate Gap}_{b,c,t} \times \text{QE Exposure}_{b,c,t-1}$	0.32***	0.33***
	(0.04)	(0.04)
N	120,166	121,500
Bank×Episode FE	Yes	Yes
County×Year FE	Yes	No
County FE	No	Yes

Table 8: ZLB and QE Deposit Exposure and Bank Deposit Beta (Bank-Quarter)

This table reports estimates of the effect of ZLB and QE exposure on bank deposit beta, the interest-rate sensitivity of deposit costs. Panel A reports OLS estimates using the bank's own ZLB exposure. ZLB Exposure$_{b,t-1}$ is the standardized twelve-quarter trailing moving average of the bank's ZLB-era deposit inflows, scaled by its total assets at the start of the ZLB episode (2008Q2 for the post-GFC episode and 2019Q4 for the post-pandemic episode), and QE Exposure$_{b,t-1}$ is the standardized twelve-quarter trailing moving average of QE-driven deposit inflows scaled by lagged assets. Panel B reports reduced-form estimates that address the potential endogeneity of a bank's own ZLB deposit exposure, replacing ZLB Exposure$_{b,t-1}$ with the shift-share instrument ZLB Exposure$^{\text{Bartik}}_{b,t-1}$ described in Table 7, which combines the bank's holdings of non-interest-bearing, savings, and time deposits at the start of the ZLB episode with national leave-one-out growth rates of each deposit type, scaled by the bank's total assets at the start of the episode. The dependent variable is the quarterly change in a deposit spread, defined as the Federal Funds Rate (FFR) minus a deposit rate: the bank's average deposit rate in column (1), its savings deposit rate in column (2), and its time deposit rate in column (3). Because the average deposit rate mixes product rates with the bank's deposit composition — a bank with more non-interest-bearing deposits mechanically has a lower average rate — the product-level spreads in columns (2)–(3) measure pricing within a deposit product. A *negative* coefficient on an interaction term implies a *higher* beta: more-exposed banks compress the spread by passing more of a rate hike into deposit rates. The sample pools the post-GFC (2016Q1–2020Q1) and post-pandemic (2022Q2–2025Q4) non-ZLB periods. For each exposure, the table reports its direct effect and its interaction with the cumulative change in the FFR over the current and preceding three quarters, $\sum_{h=0}^{3} \Delta\text{FFR}_{t-h}$ (the reported interaction coefficient is the sum of the current and three lagged interaction terms). All specifications include two sets of controls, indicated by the Controls row, each interacted with the current and three lagged FFR changes: bank-level controls, namely lagged log total assets and bank-level deposit HHI (measured in 2008 for the post-GFC episode and 2019 for the post-pandemic episode), following Drechsler, Savov and Schnabl (2017); and composition controls, namely the bank's holdings of non-interest-bearing, savings, and time deposits at the start of the ZLB episode (2008Q2 or 2019Q4), each scaled by total assets at that date. Because the instrument in Panel B is built from the same holdings as the composition controls, these controls absorb any direct effect of a bank's deposit mix on its deposit beta. The direct effects of the HHI and composition variables are absorbed by the bank×episode fixed effects, and the direct effects of the FFR changes by the quarter fixed effects. R^2 is adjusted. All non-standardized variables are winsorized at the 1st and 99th percentiles. Robust standard errors, clustered two-way by bank and state-quarter, are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: OLS, Own ZLB Exposure

Dependent Variable:	(1) ΔDep Spread$_{b,t}$	(2) ΔSavings Dep Spread$_{b,t}$	(3) ΔTime Dep Spread$_{b,t}$
ZLB Exposure$_{b,t-1}$	0.000***	0.000***	0.000***
	(0.000)	(0.000)	(0.000)
QE Exposure$_{b,t-1}$	0.000*	0.000	0.000***
	(0.000)	(0.000)	(0.000)
ZLB Exposure$_{b,t-1}\times$ $\sum_{h=0}^{3}\Delta\text{FFR}_{t-h}$	-0.019***	-0.034***	-0.010***
	(0.003)	(0.004)	(0.003)
QE Exposure$_{b,t-1}\times$ $\sum_{h=0}^{3}\Delta\text{FFR}_{t-h}$	-0.005***	-0.003	-0.005*
	(0.002)	(0.003)	(0.003)
N	148,443	147,088	147,451
R^2	0.94	0.85	0.79
Controls	Yes	Yes	Yes
Bank×Episode FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes

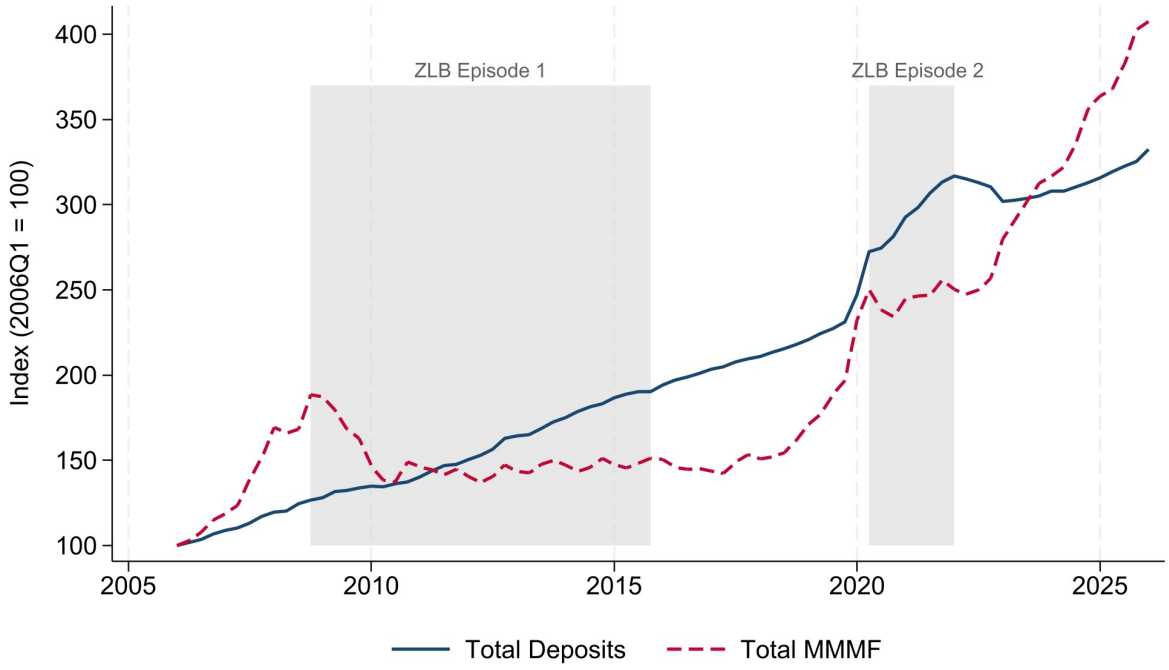
Table 8 continued from previous page

Panel B: Reduced Form, Shift-Share ZLB Exposure Instrument

Dependent Variable:	(1) ΔDep Spread$_{b,t}$	(2) ΔSavings Dep Spread$_{b,t}$	(3) ΔTime Dep Spread$_{b,t}$
ZLB Exposure$^{\text{Bartik}}_{b,t-1}$	0.000***	-0.000	0.000**
	(0.000)	(0.000)	(0.000)
QE Exposure$_{b,t-1}$	0.000	-0.000	0.000**
	(0.000)	(0.000)	(0.000)
ZLB Exposure$^{\text{Bartik}}_{b,t-1} \times$	-0.024***	-0.007	-0.017**
$\sum_{h=0}^{3} \Delta\text{FFR}_{t-h}$	(0.003)	(0.005)	(0.007)
QE Exposure$_{b,t-1} \times$	-0.004**	-0.002	-0.004*
$\sum_{h=0}^{3} \Delta\text{FFR}_{t-h}$	(0.002)	(0.003)	(0.003)
N	148,443	147,088	147,451
R^2	0.94	0.85	0.79
Controls	Yes	Yes	Yes
Bank$\times$Episode FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes

Figure 1: Total U.S. Bank Deposits vs. Money Market Mutual Funds

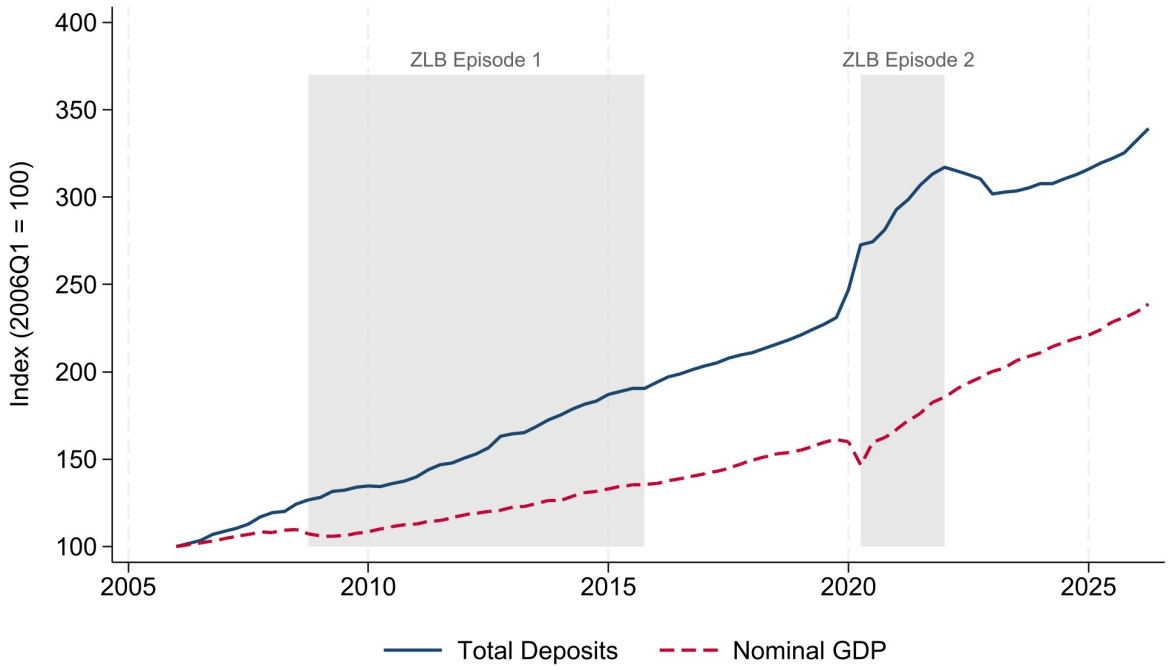
This figure plots total deposits at U.S. commercial banks against total money market mutual fund (MMMF) assets from 2006Q1 through 2026Q1, with both series indexed to 100 in 2006Q1. Bank deposits and MMMF assets tend to move inversely as depositors substitute between insured bank deposits and money market funds in response to the rate environment: MMMF assets surge during rate-hiking cycles, when money funds pass through higher yields than banks, and recede when rates fall.



Source: Federal Reserve Economic Data (FRED), series DPSACBW027SBOG (deposits, all commercial banks) and MMMFFAQ027S (money market funds, total financial assets). Both series are indexed to 100 in 2006Q1.

Figure 2: Total U.S. Bank Deposits vs. Nominal GDP

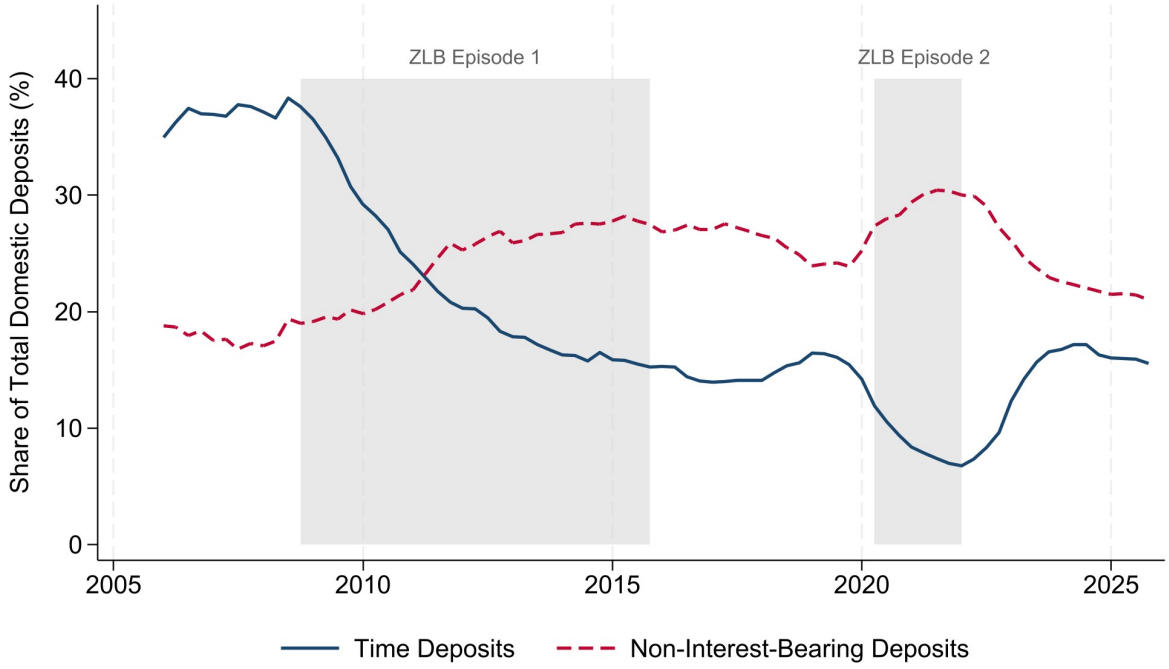
This figure plots total deposits at U.S. commercial banks against nominal gross domestic product from 2006Q1 through 2026Q1, with both series indexed to 100 in 2006Q1. Bank deposits have grown substantially faster than nominal output over the sample period, reflecting the expansion of the deposit base relative to the broader economy.



Source: Federal Reserve Economic Data (FRED), series DPSACBW027SBOG (deposits, all commercial banks) and GDP (gross domestic product). Both series are indexed to 100 in 2006Q1.

Figure 3: Deposit Composition Over Time: Time and Non-Interest-Bearing Deposit Shares

This figure plots the composition of domestic deposits in the U.S. banking system from 2006Q1 through 2025Q4. The solid line is the share of total domestic deposits held as time deposits, and the dashed line is the share held as non-interest-bearing deposits, each computed from aggregate FDIC industry balances. The figure documents a pronounced compositional shift during the zero-lower-bound episodes: the time-deposit share declines sharply while the non-interest-bearing share rises, as depositors abandon low-yielding time deposits and reallocate funds into liquid, non-interest-bearing accounts held for safety and liquidity.



Source: Federal Reserve Economic Data (FRED), series QBPBSTLKDPDOFFDPTDP (time deposits), QBPBSTLKDPDOFFDPNIDP (non-interest-bearing deposits), and QBPBSTLKDPDOFFDP (total domestic office deposits), author's calculations.

Figure 4: Savings Deposit Rates Across Banks: Median and Interquartile Range

This figure plots the cross-sectional distribution of annualized savings deposit rates across U.S. commercial banks from 2006Q1 through 2025Q4. The solid line is the median savings deposit rate across banks in each quarter, and the shaded region spans the 25th to 75th percentiles. Each bank's savings deposit rate is computed as its annualized interest expense on savings deposits divided by average savings deposits, from the quarterly Call Reports, with all banks weighted equally. Both the level and the dispersion of savings deposit rates track the policy cycle: rates compress toward zero during the two zero-lower-bound episodes and rise during the intervening tightening cycles, while the interquartile range widens as banks differ in the degree to which they pass higher policy rates through to depositors.

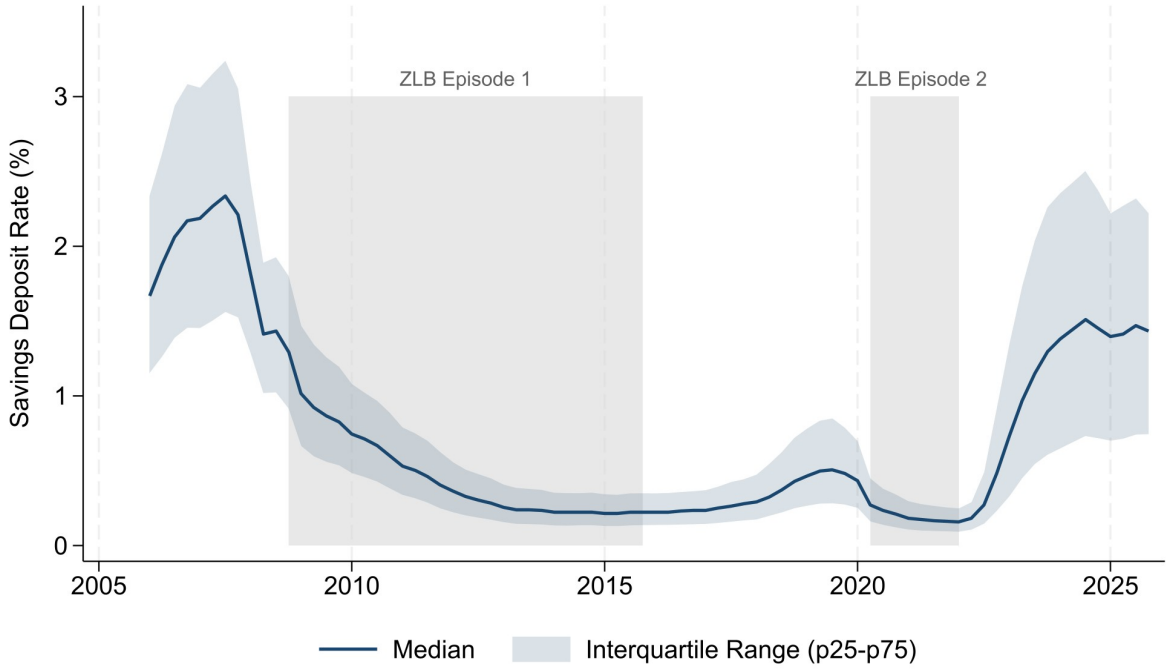


Figure 5: The Rate Gap Across Banks: Median and Interquartile Range

This figure plots the cross-sectional distribution of the deposit Rate Gap across U.S. commercial banks from 2006Q1 through 2025Q4. The Rate Gap, following Begenau and Stafford (2022), is the difference between a bank's own annualized deposit rate and the deposit-share-weighted rate of competing banks in the counties in which it operates, using predetermined deposit weights, aggregated to the bank level across its counties. To isolate cross-sectional variation, the Rate Gap is demeaned within each quarter in this figure only — that is, we subtract the cross-bank average rate gap in each quarter — so that the figure displays each bank's Rate-Gap position relative to the average bank in that quarter, the variation absorbed by the quarter fixed effects in the regressions. In the regressions themselves, we use the Rate Gap in levels, following Begenau and Stafford (2022), since the quarter fixed effects absorb the common time-series component. The solid line is the median demeaned Rate Gap across banks in each quarter, and the shaded region spans the 25th to 75th percentiles. The dispersion of the Rate Gap tracks the policy cycle: the interquartile range compresses during the two zero-lower-bound episodes, when all banks' deposit rates are pinned near zero and cross-bank rate differences vanish, and widens during the intervening tightening cycles as banks differ in the degree to which they pass higher policy rates through to depositors.

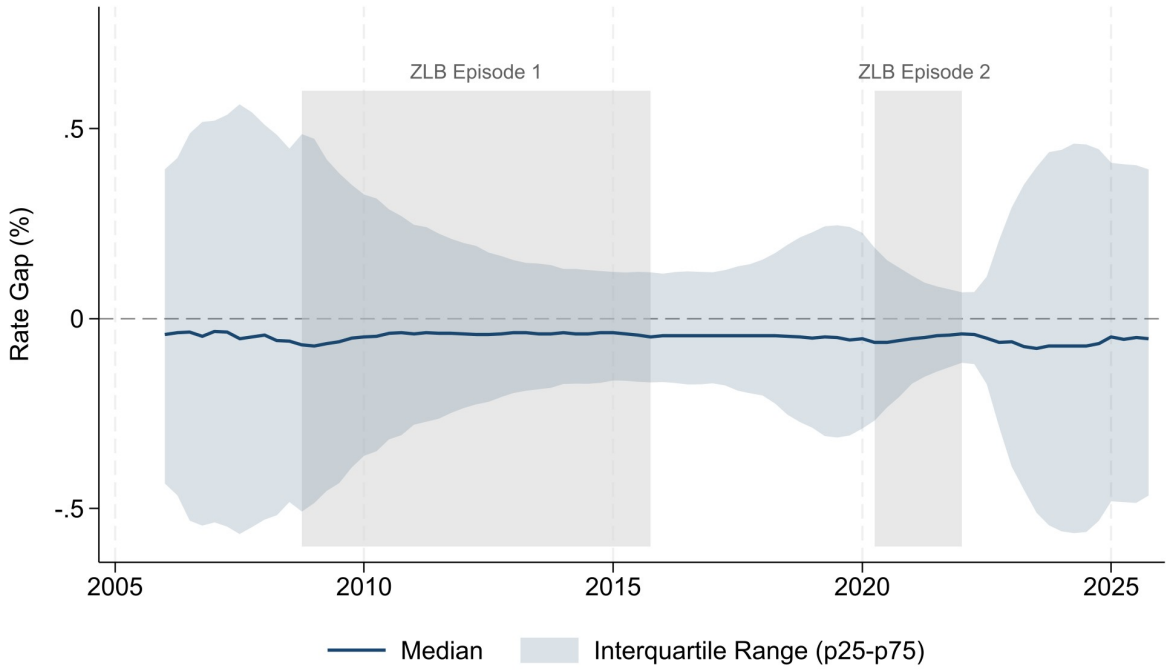
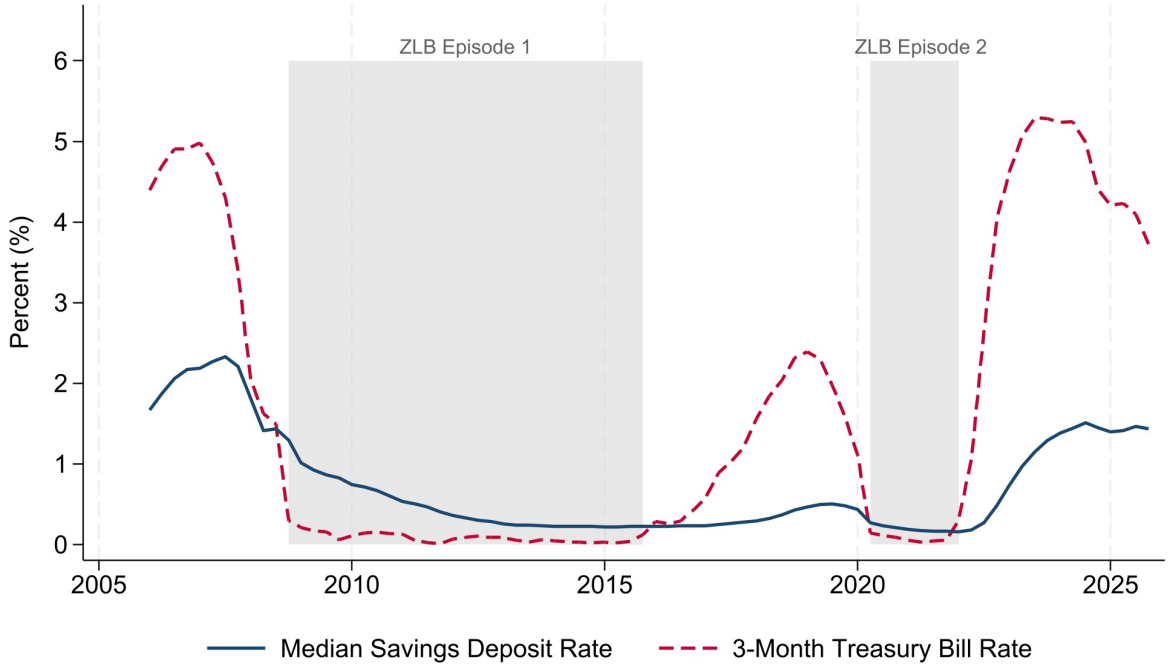


Figure 6: Median Savings Deposit Rate vs. the 3-Month Treasury Bill Rate

This figure plots the median savings deposit rate across U.S. commercial banks against the 3-month Treasury bill rate from 2006Q1 through 2025Q4. The solid line is the cross-bank median of the annualized savings deposit rate, computed from the quarterly Call Reports as interest expense on savings deposits divided by average savings deposits, with all banks weighted equally; the dashed line is the quarterly average of the 3-month Treasury bill secondary-market rate. We use the 3-month Treasury bill rate as a proxy for the product rates of money market funds — a key outside option for depositors — because it closely tracks money market fund yields while offering the long sample coverage that the money market fund rate series available on FRED do not.



Source: Federal Reserve Economic Data (FRED), series TB3MS (3-month Treasury bill secondary market rate, discount basis), and the author's calculations from the quarterly Call Reports.

Appendix

Unconventional Monetary Policy and Deposit Retention

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Table A.1: Variable Definitions

This table provides definitions for the variables and their corresponding data terms used in this paper.

Variable	Definition	Data Terms
$\text{Assets}_{b,t}$	Total assets of bank b at quarter t.	$\max(\text{RCFD2170}_{b,t}, \text{RCON2170}_{b,t})$
$\text{Reserves}_{b,t}$	Total reserves of bank b at quarter t.	$\max(\text{RCFD0090}_{b,t}, \text{RCON0090}_{b,t})$
$\text{Total Dep}_{b,t}$	Total deposits of bank b at quarter t.	$\text{RCON2200}_{b,t} + \text{RCFN2200}_{b,t}$
$\text{Total Dep}_{b,c,t}$	Total deposits across all branches of bank b located in county c as of June 30 of year t.	$\sum_{br \in bc} \text{Depsumbr}_{br,t}$
$\text{Insured Dep}_{b,t}$	Total insured deposits of bank b at quarter t.	$\text{RCONF045}_{b,t} + \text{RCONF049}_{b,t}$
$\text{Uninsured Dep}_{b,t}$	Total uninsured deposits of bank b at quarter t.	$\text{RCONF051}_{b,t} + \text{RCONF047}_{b,t} + \text{RCFN2200}_{b,t}$
$\text{ZLB Exposure}_{b,t}$	Standardized twelve-quarter trailing moving average of bank b's total deposit change, counting only ZLB quarters (non-ZLB quarters contribute zero), over the window from $t-12$ to $t-1$, scaled by the bank's total assets in the quarter q_e before the ZLB episode (2008Q2 for Episode 1; 2019Q4 for Episode 2).	$\text{STD}\left(\frac{1}{12}\sum_{k=1}^{12}\frac{\Delta\text{Dep}_{b,t-k}\cdot\mathbb{1}[\text{ZLB}_{t-k}]}{\text{Assets}_{b,q_e}}\right)$
$\text{ZLB Exposure}_{b,c,t}$	Standardized three-year trailing moving average of bank b's total deposit change in county c, counting only ZLB years (non-ZLB years contribute zero), over the window from $t-3$ to $t-1$, scaled by the bank's total assets at the start of the ZLB episode, s_e (SOD year 2008 for Episode 1; 2019 for Episode 2).	$\text{STD}\left(\frac{1}{3}\sum_{k=1}^{3}\frac{\Delta\text{Dep}_{b,c,t-k}\cdot\mathbb{1}[\text{ZLB}_{t-k}]}{\text{Assets}_{b,s_e}}\right)$

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Table A.1 continued from previous page

Variable	Definition	Data Terms
$G_{-b,j,\tau}$	National leave-one-out growth rate of deposit type j in quarter τ: the change in type-j deposits summed over all banks other than b, divided by their lagged type-j deposits, computed over banks with type-j deposits observed in both τ and $\tau-1$, excluding bank-quarters affected by mergers and acquisitions.	$\frac{\sum_{b' \neq b} \Delta \text{Dep}_{b',j,\tau}}{\sum_{b' \neq b} \text{Dep}_{b',j,\tau-1}}$
$\text{ZLB Exposure}_{b,t}^{\text{Bartik}}$	Standardized shift-share (Bartik) instrument for ZLB Exposure: the twelve-quarter trailing moving average, over the window from $t-12$ to $t-1$, of bank b's predicted deposit inflow, counting only ZLB quarters (non-ZLB quarters contribute zero). The predicted inflow multiplies the bank's holdings of non-interest-bearing (NI), savings (S), and time (T) deposits in the quarter q_e before the ZLB episode (2008Q2 for Episode 1; 2019Q4 for Episode 2), each scaled by the bank's total assets in that quarter, by the national leave-one-out growth rate of the corresponding deposit type, $G_{-b,j,t-k}$, and sums across the three types.	$\text{STD}\left(\frac{1}{12}\sum_{k=1}^{12} \mathbb{1}[\text{ZLB}_{t-k}] \sum_{j \in \{\text{NI,S,T}\}} \frac{\text{Dep}_{b,j,q_e}}{\text{Assets}_{b,q_e}} G_{-b,j,t-k}\right)$
$\text{MBS}_{b,t}$	Total mortgage-backed securities of bank b at quarter t.	

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Table A.1 continued from previous page

Variable	Definition	Data Terms
	$\max(\text{RCFD1698}, \text{RCON1698}) + \max(\text{RCFD1701}, \text{RCON1701})$	
	$+ \max(\text{RCFD1703}, \text{RCON1703}) + \max(\text{RCFD1706}, \text{RCON1706})$	
	$+ \max(\text{RCFD1709}, \text{RCON1709}) + \max(\text{RCFD1711}, \text{RCON1711})$	
	$+ \max(\text{RCFD1714}, \text{RCON1714}) + \max(\text{RCFD1716}, \text{RCON1716})$	
	$+ \max(\text{RCFD1718}, \text{RCON1718}) + \max(\text{RCFD1731}, \text{RCON1731})$	
	$+ \max(\text{RCFD1733}, \text{RCON1733}) + \max(\text{RCFD1735}, \text{RCON1735})$	*If* $t < 2009Q2$
	$\max(\text{RCFDG300}, \text{RCONG300}) + \max(\text{RCFDG302}, \text{RCONG302})$	
	$+ \max(\text{RCFDG304}, \text{RCONG304}) + \max(\text{RCFDG306}, \text{RCONG306})$	
	$+ \max(\text{RCFDG308}, \text{RCONG308}) + \max(\text{RCFDG310}, \text{RCONG310})$	
	$+ \max(\text{RCFDG312}, \text{RCONG312}) + \max(\text{RCFDG314}, \text{RCONG314})$	
	$+ \max(\text{RCFDG316}, \text{RCONG316}) + \max(\text{RCFDG318}, \text{RCONG318})$	
	$+ \max(\text{RCFDG320}, \text{RCONG320}) + \max(\text{RCFDG322}, \text{RCONG322})$	
	$+ \max(\text{RCFDG324}, \text{RCONG324}) + \max(\text{RCFDG326}, \text{RCONG326})$	
	$+ \max(\text{RCFDG328}, \text{RCONG328}) + \max(\text{RCFDG330}, \text{RCONG330})$	*If* $2009Q2 \leq t < 2010Q4$
	$\max(\text{RCFDG300}, \text{RCONG300}) + \max(\text{RCFDG302}, \text{RCONG302})$	
	$+ \max(\text{RCFDG304}, \text{RCONG304}) + \max(\text{RCFDG306}, \text{RCONG306})$	
	$+ \max(\text{RCFDG308}, \text{RCONG308}) + \max(\text{RCFDG310}, \text{RCONG310})$	
	$+ \max(\text{RCFDG312}, \text{RCONG312}) + \max(\text{RCFDG314}, \text{RCONG314})$	
	$+ \max(\text{RCFDG316}, \text{RCONG316}) + \max(\text{RCFDG318}, \text{RCONG318})$	
	$+ \max(\text{RCFDG320}, \text{RCONG320}) + \max(\text{RCFDG322}, \text{RCONG322})$	
	$+ \max(\text{RCFDK142}, \text{RCONK142}) + \max(\text{RCFDK144}, \text{RCONK144})$	
	$+ \max(\text{RCFDK146}, \text{RCONK146}) + \max(\text{RCFDK148}, \text{RCONK148})$	
	$+ \max(\text{RCFDK150}, \text{RCONK150}) + \max(\text{RCFDK152}, \text{RCONK152})$	
	$+ \max(\text{RCFDK154}, \text{RCONK154}) + \max(\text{RCFDK156}, \text{RCONK156})$	*If* $2010Q4 \leq t < 2018Q1$
	$\max(\text{RCFDG300}, \text{RCONG300}) + \max(\text{RCFDG302}, \text{RCONG302})$	
	$+ \max(\text{RCFDG304}, \text{RCONG304}) + \max(\text{RCFDG306}, \text{RCONG306})$	
	$+ \max(\text{RCFDG308}, \text{RCONG308}) + \max(\text{RCFDG310}, \text{RCONG310})$	
	$+ \max(\text{RCFDG312}, \text{RCONG312}) + \max(\text{RCFDG314}, \text{RCONG314})$	
	$+ \max(\text{RCFDG316}, \text{RCONG316}) + \max(\text{RCFDG318}, \text{RCONG318})$	
	$+ \max(\text{RCFDG320}, \text{RCONG320}) + \max(\text{RCFDG322}, \text{RCONG322})$	
	$+ \max(\text{RCFDK142}, \text{RCONK142}) + \max(\text{RCFDK144}, \text{RCONK144})$	
	$+ \max(\text{RCFDK146}, \text{RCONK146}) + \max(\text{RCFDK148}, \text{RCONK148})$	
	$+ \max(\text{RCFDK150}, \text{RCONK150}) + \max(\text{RCFDK152}, \text{RCONK152})$	
	$+ \max(\text{RCFDK154}, \text{RCONK154}) + \max(\text{RCFDK156}, \text{RCONK156})$	
	$+ \text{RCONHT54} + \text{RCONHT56}$	*If* $t \geq 2018Q1$

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Table A.1 continued from previous page

Variable	Definition	Data Terms
$\text{Rate Gap}_{b,t}$	Bank b's deposit-footprint-weighted average of its bank-county rate gaps across the counties in which it operates, using predetermined deposit weights.	$\sum_c \omega_{b,c}\,\text{Rate Gap}_{b,c,t}$
$\text{Rate Gap}_{b,c,t}$	Difference between bank b's own deposit rate and the deposit-share-weighted average deposit rate of competing banks in county c, using predetermined deposit-share weights and excluding bank b itself. Following Begenau and Stafford (2022).	$\text{Dep Rate}_{b,t} - \sum_{b' \neq b} s_{b',c}\,\text{Dep Rate}_{b',t}$
$\text{Rate Gap}^{\text{Sav}}_{b,c,t}$	Difference between bank b's own savings deposit rate and the deposit-share-weighted average savings deposit rate of competing banks in county c, using predetermined deposit-share weights and excluding bank b itself. Constructed as $\text{Rate Gap}_{b,c,t}$ but using the savings deposit rate, to isolate competitive pricing from deposit composition.	$\text{Sav Rate}_{b,t} - \sum_{b' \neq b} s_{b',c}\,\text{Sav Rate}_{b',t}$
$\text{QE Exposure}_{b,t}$	Standardized twelve-quarter trailing moving average of bank b's QE-driven deposit inflows scaled by lagged assets, over the window from $t-12$ to $t-1$. Following Yang (2026).	$\text{STD}\left(\frac{1}{12}\sum_{k=1}^{12}\frac{Z_{b,t-k}}{\text{Assets}_{b,t-k-1}}\right)$
$\text{QE Exposure}_{b,c,t}$	Standardized three-year trailing moving average of bank b's QE-driven deposit inflows in county c scaled by lagged assets, over the window from $t-3$ to $t-1$. Following Yang (2026).	$\text{STD}\left(\frac{1}{3}\sum_{k=1}^{3}\frac{Z_{b,c,t-k}}{\text{Assets}_{b,t-k-1}}\right)$

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Table A.1 continued from previous page

Variable	Definition	Data Terms
Dep Spread$_{b,t}$	Difference between the federal funds rate and bank b's annualized deposit rate at quarter t, where the deposit rate is interest expense on deposits divided by average deposits.	FFR$_t$ − Dep Rate$_{b,t}$
Savings Dep Spread$_{b,t}$	Difference between the federal funds rate and bank b's annualized savings deposit rate at quarter t, where the savings deposit rate is interest expense on savings deposits (including money market deposit accounts) divided by average savings deposits.	FFR$_t$ − Savings Dep Rate$_{b,t}$
Time Dep Spread$_{b,t}$	Difference between the federal funds rate and bank b's annualized time deposit rate at quarter t, where the time deposit rate is interest expense on time deposits divided by average time deposits.	FFR$_t$ − Time Dep Rate$_{b,t}$

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Table A.1 continued from previous page

Variable	Definition	Data Terms
$Z_{c,t}$	County-level proxy for QE-generated deposit creation in county c at quarter t. Each domestic primary dealer's net Fed security purchases in quarter t ($\text{Net Purchase}_{pd,t}$) are allocated across the counties in which the dealer's parent bank holding company operates, in proportion to the county's share of that dealer's business, then summed across all dealers operating in county c. Here PD_c is the set of primary dealers with a business presence in county c; $\text{Business}_{pd,c}$ is dealer pd's predetermined (2008) business activity in county c, measured by the deposits of its parent bank holding company's branches in that county (from the FDIC Summary of Deposits); and $\text{Business}_{pd} = \sum_c \text{Business}_{pd,c}$ is the dealer's total business across all counties, so that $\text{Business}_{pd,c}/\text{Business}_{pd}$ is the predetermined share of dealer pd's footprint located in county c. Following Yang (2026).	$\sum_{pd \in PD_c} \frac{\text{Business}_{pd,c}}{\text{Business}_{pd}} \times \text{Net Purchase}_{pd,t}$
$Z_{b,c,t}$	Bank-county-level QE deposit inflow: the county-level measure $Z_{c,t}$ allocated to bank b according to its predetermined (2008) share of county-c deposits. Captures within-bank, across-county variation in QE exposure. Following Yang (2026).	$\frac{\text{Deposits}_{b,c}}{\text{Deposits}_{c}} \times Z_{c,t}$

continued on next page

Table A.1 continued from previous page

Variable	Definition	Data Terms
$Z_{b,t}$	Bank-level QE deposit inflow: bank b's bank-county inflows $Z_{b,c,t}$ aggregated across all counties in which it maintains a physical presence. Banks that are subsidiaries of primary dealers are excluded. Following Yang (2026).	$\sum_{c \in B_c} Z_{b,c,t}$
$Z_{\text{mbs},b,t}$	Bank b's ex-ante share of aggregate MBS holdings, interacted with the Federal Reserve's net MBS purchases at quarter t. This shift-share instrument proxies for the bank's structural capacity to participate in the direct asset-swap channel of QE, following Rodnyansky and Darmouni (2017) and Yang (2026).	$\frac{\text{MBS}_{b,t-1}}{\text{MBS}_{t-1}} \times \text{Net MBS Purchase}_t$
$Z_{b,t}^{R1}$	Reserve-based shift-share exposure to QE, following Acharya et al. (2024): aggregate reserve growth at quarter t interacted with bank b's predetermined reserve intensity, its trailing four-quarter average share of aggregate reserves.	$\Delta \ln(\text{Reserves}_t) \times \frac{1}{4} \sum_{k=1}^{4} \frac{\text{Reserves}_{b,t-k}}{\text{Reserves}_{t-k}}$

Table A.2: Summary Statistics

This table reports summary statistics for the primary variables used in this study. All variables, with the exception of those in logarithmic or standardized form, are winsorized at the 1st and 99th percentiles.

	Mean	SD	Min	Max	N
$Z_{b,t}^{R1}$	0.000	0.002	-0.011	0.012	114,910
$Z_{\text{mbs},b,t}/\text{Assets}_{b,t-1}$	0.002	0.003	0.000	0.015	392,916
$Z_{b,t}/\text{Assets}_{b,t-1}$	0.001	0.004	0.000	0.025	392,917
$Z_{b,c,t}/\text{Assets}_{b,t-1}$	0.001	0.005	0.000	0.037	338,647
$\Delta\text{Dep}_{b,t}/\text{Assets}_{b,t-1}$	0.011	0.042	-0.236	0.316	392,917
$\Delta\text{Dep}_{b,c,t}/\text{Assets}_{b,t-1}$	0.011	0.037	-0.092	0.189	332,469
$\Delta\text{Uninsured Dep}_{b,t}/\text{Assets}_{b,t-1}$	0.004	0.041	-0.166	0.139	392,917
$\Delta\text{Insured Dep}_{b,t}/\text{Assets}_{b,t-1}$	0.007	0.033	-0.070	0.178	392,917
$\text{Rate Gap}_{b,t}$	0.003	0.006	-0.012	0.019	629,618
$\text{Rate Gap}_{b,c,t}$	0.002	0.005	-0.012	0.019	474,225
$\text{Rate Gap}_{b,c,t}^{\text{Sav}}$	0.001	0.006	-0.014	0.021	472,709
$\text{QE Exposure}_{b,t}$	0.000	1.000	-0.367	6.115	622,419
$\text{QE Exposure}_{b,c,t}$	0.000	1.000	-0.258	6.913	427,252
ZLB_t	0.374	0.484	0.000	1.000	632,999
SVB_t	0.015	0.120	0.000	1.000	632,999
$\text{ZLB Exposure}_{b,t}$	0.000	1.000	-1.869	7.020	395,685
$\text{ZLB Exposure}_{b,t}^{\text{Bartik}}$	0.000	1.000	-4.033	3.900	151,396
$\text{ZLB Exposure}_{b,c,t}$	0.000	1.000	-1.814	8.343	242,518
$\Delta\text{Dep Spread}_{b,t}$	0.000	0.004	-0.013	0.013	384,939
$\Delta\text{Savings Dep Spread}_{b,t}$	0.000	0.005	-0.014	0.013	380,784
$\Delta\text{Time Dep Spread}_{b,t}$	0.000	0.005	-0.014	0.013	382,311
ΔFFR_t	-0.000	0.005	-0.016	0.014	665,801

Table A.3: Robustness of the ZLB and QE Effects to Small-Market Rate Gaps (Bank-Quarter)

This table reproduces Table 3 using an alternative Rate Gap constructed only from "small" markets — counties in which the bank raises less than half of its total deposits. Because a bank sets its uniform deposit rate based on conditions across all of its markets, the Rate Gap in the bank's dominant markets may be influenced by the bank's own choices; restricting to markets where the bank raises less than half of its deposits isolates the competing markets least likely to affect those choices, sharpening the exogeneity of the Rate Gap. The measure is defined only for banks operating in at least two counties. Panels A, B, and C split the dependent variable into total, uninsured, and insured deposits, respectively. Variable definitions and specifications follow Table 3. All specifications include bank×regime and quarter fixed effects, where the regime denotes the ZLB versus non-ZLB monetary environment. The main effects of ZLB_t and SVB_t are absorbed. Robust standard errors, clustered two-way by bank and quarter, are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Total Deposits

	(1)	(2)	(3)	(4)
Dependent Variable:	$\Delta\text{Dep}_{b,t} / \text{Assets}_{b,t-1}$			
$\text{Rate Gap}_{b,t}$	0.46***	0.46***	0.49***	0.47***
	(0.07)	(0.07)	(0.06)	(0.07)
$\text{QE Exposure}_{b,t-1}$		-0.000	-0.000	-0.000
		(0.000)	(0.000)	(0.000)
$\text{Rate Gap}_{b,t} \times \text{ZLB}_t$		0.11	0.01	0.03
		(0.28)	(0.24)	(0.24)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1}$		0.06	0.22***	0.23**
		(0.08)	(0.07)	(0.11)
$\text{QE Exposure}_{b,t-1} \times \text{ZLB}_t$		0.002***	0.003***	0.003***
		(0.001)	(0.001)	(0.001)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1} \times \text{ZLB}_t$			-0.71***	-0.71***
			(0.14)	(0.16)
$\text{Rate Gap}_{b,t} \times \text{SVB}_t$				0.78***
				(0.16)
$\text{QE Exposure}_{b,t-1} \times \text{SVB}_t$				-0.001
				(0.001)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1} \times \text{SVB}_t$				-0.22**
				(0.09)
N	320,905	317,275	317,275	317,275
Bank×Regime FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes

Table A.3 continued from previous page

Panel B: Uninsured Deposits

	(1)	(2)	(3)	(4)
Dependent Variable:		ΔUninsured $\text{Dep}_{b,t}/\text{Assets}_{b,t-1}$		
$\text{Rate Gap}_{b,t}$		0.11***	0.12***	0.13***
		(0.04)	(0.04)	(0.04)
$\text{QE Exposure}_{b,t-1}$		-0.001*	-0.001*	-0.000
		(0.000)	(0.000)	(0.000)
$\text{Rate Gap}_{b,t} \times \text{ZLB}_t$		0.03	-0.01	-0.02
		(0.23)	(0.21)	(0.21)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1}$		0.04	0.10***	0.11**
		(0.03)	(0.03)	(0.05)
$\text{QE Exposure}_{b,t-1} \times \text{ZLB}_t$		0.002***	0.002***	0.002***
		(0.001)	(0.001)	(0.001)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1} \times \text{ZLB}_t$			-0.28***	-0.29***
			(0.08)	(0.09)
$\text{Rate Gap}_{b,t} \times \text{SVB}_t$				-0.21**
				(0.10)
$\text{QE Exposure}_{b,t-1} \times \text{SVB}_t$				-0.001**
				(0.001)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1} \times \text{SVB}_t$				-0.00
				(0.05)
N	320,905	317,275	317,275	317,275
Bank×Regime FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes

Panel C: Insured Deposits

	(1)	(2)	(3)	(4)
Dependent Variable:		ΔInsured $\text{Dep}_{b,t}/\text{Assets}_{b,t-1}$		
$\text{Rate Gap}_{b,t}$		0.35***	0.37***	0.34***
		(0.05)	(0.05)	(0.05)
$\text{QE Exposure}_{b,t-1}$		0.000	0.000	0.000
		(0.000)	(0.000)	(0.000)
$\text{Rate Gap}_{b,t} \times \text{ZLB}_t$		0.08	0.02	0.05
		(0.19)	(0.18)	(0.18)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1}$		0.02	0.12**	0.11*
		(0.05)	(0.05)	(0.07)
$\text{QE Exposure}_{b,t-1} \times \text{ZLB}_t$		0.001	0.001***	0.001***
		(0.001)	(0.000)	(0.000)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1} \times \text{ZLB}_t$			-0.43***	-0.42***
			(0.09)	(0.10)
$\text{Rate Gap}_{b,t} \times \text{SVB}_t$				0.99***
				(0.21)
$\text{QE Exposure}_{b,t-1} \times \text{SVB}_t$				0.001
				(0.000)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1} \times \text{SVB}_t$				-0.22***
				(0.07)
N	320,905	317,275	317,275	317,275
Bank×Regime FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes

Table A.4: Separating the Effects of the ZLB and QE on Local Deposit-Flow Sensitivity to the Rate Gap: Savings Deposit Rate (Bank-County-Year)

This table reproduces Table 5, replacing the Rate Gap based on the overall deposit rate with one based on the savings deposit rate only. The overall deposit rate reflects a bank's deposit composition—a bank funded largely by non-interest-bearing demand deposits mechanically exhibits a lower deposit rate, and hence a more negative Rate Gap, even if it prices each deposit type competitively—so the overall Rate Gap may capture composition rather than competitive pricing. Using the savings deposit rate, the largest interest-bearing category, isolates competitive pricing from composition. The Rate Gap is otherwise constructed as in Table 5: the difference between the bank's own annualized savings deposit rate and the deposit-share-weighted savings deposit rate of competing banks in the county, using predetermined deposit weights. The dependent variable is the annual change in bank-county deposits, from the FDIC Summary of Deposits, scaled by lagged bank assets. ZLB_t equals one when the Federal Funds rate is at its effective floor during the SOD year. SVB_t equals one in the 2023 SOD year, capturing the deposit disruption around the failure of Silicon Valley Bank. $\text{QE Exposure}_{b,c,t-1}$ is a standardized three-year lagged moving average of QE-driven deposit inflows scaled by lagged assets, following Yang (2026). Column (1) is the baseline; Column (2) adds the ZLB and QE-Exposure interactions; Column (3) adds the triple interaction; Column (4) adds the SVB interactions and the triple interaction. All specifications include bank×year, county×year, and bank×county fixed effects; the main effects of ZLB_t and SVB_t are absorbed. Robust standard errors, clustered two-way by bank and county, are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)	(4)
Dependent Variable:	$\Delta\text{Dep}_{b,c,t} / \text{Assets}_{b,t-1}$			
$\text{Rate Gap}^{\text{Sav}}_{b,c,t}$	0.91***	0.92***	0.93***	0.74***
	(0.10)	(0.11)	(0.11)	(0.11)
$\text{QE Exposure}_{b,c,t-1}$		-0.003***	-0.004***	-0.002***
		(0.000)	(0.000)	(0.000)
$\text{Rate Gap}^{\text{Sav}}_{b,c,t} \times \text{ZLB}_t$		-0.09	-0.10	0.07
		(0.29)	(0.29)	(0.30)
$\text{Rate Gap}^{\text{Sav}}_{b,c,t} \times \text{QE Exposure}_{b,c,t-1}$		0.07**	0.10***	0.07
		(0.03)	(0.04)	(0.04)
$\text{QE Exposure}_{b,c,t-1} \times \text{ZLB}_t$		0.002***	0.002***	0.000
		(0.000)	(0.000)	(0.000)
$\text{Rate Gap}^{\text{Sav}}_{b,c,t} \times \text{QE Exposure}_{b,c,t-1} \times \text{ZLB}_t$			-0.21**	-0.18**
			(0.08)	(0.08)
$\text{Rate Gap}^{\text{Sav}}_{b,c,t} \times \text{SVB}_t$				0.74***
				(0.21)
$\text{QE Exposure}_{b,c,t-1} \times \text{SVB}_t$				-0.003***
				(0.001)
$\text{Rate Gap}^{\text{Sav}}_{b,c,t} \times \text{QE Exposure}_{b,c,t-1} \times \text{SVB}_t$				0.06
				(0.05)
N	366,157	331,904	331,904	331,904
Bank×Year FE	Yes	Yes	Yes	Yes
County×Year FE	Yes	Yes	Yes	Yes
Bank×County FE	Yes	Yes	Yes	Yes

Table A.5: Cross-Bank Correlations of Policy Exposure and Deposit Composition at the End of Each ZLB Episode

This table reports cross-bank (Pearson) correlations among policy-exposure measures and deposit composition, computed on a single cross-section at the end of each zero-lower-bound (ZLB) episode: 2015Q4 for Episode 1 and 2022Q1 for Episode 2, the last ZLB quarter before each rate-normalization cycle. ZLB Exposure is the twelve-quarter trailing average of the bank's ZLB-era deposit inflows scaled by its total assets at the start of the episode (2008Q2 or 2019Q4), and ZLB Exposure^{Bartik} is its shift-share instrument (Table 7). The table combines both with QE Exposure and the shares of the bank's domestic deposits held in time deposits, in non-interest-bearing accounts, and in uninsured accounts. Columns are numbered to match the row labels.

Episode 1 (2015Q4, End of Post-GFC ZLB)

	(1)	(2)	(3)	(4)	(5)	(6)
(1) ZLB Exposure	1.00					
(2) ZLB Exposure^{Bartik}	0.11	1.00				
(3) QE Exposure	0.01	0.14	1.00			
(4) Time Deposit Share	-0.03	-0.60	-0.01	1.00		
(5) Non-Interest-Bearing Share	0.05	0.37	0.08	-0.50	1.00	
(6) Uninsured Share	0.33	0.29	0.20	-0.36	0.31	1.00

Episode 2 (2022Q1, End of Pandemic ZLB)

	(1)	(2)	(3)	(4)	(5)	(6)
(1) ZLB Exposure	1.00					
(2) ZLB Exposure^{Bartik}	0.34	1.00				
(3) QE Exposure	0.03	0.09	1.00			
(4) Time Deposit Share	-0.29	-0.76	-0.03	1.00		
(5) Non-Interest-Bearing Share	0.22	0.59	0.08	-0.47	1.00	
(6) Uninsured Share	0.27	0.37	0.16	-0.32	0.35	1.00

Table A.6: Cross-Bank Regressions of Deposit Composition on Policy Exposure

This table reports cross-bank regressions that flesh out the correlations in Table A.5. The dependent variable is the share of the bank's domestic deposits held in uninsured accounts (column 1), non-interest-bearing accounts (column 2), and time deposits (column 3), measured at the end of each zero-lower-bound (ZLB) episode: 2015Q4 for Episode 1 and 2022Q1 for Episode 2. Each regression stacks the two end-of-episode cross-sections and includes an episode fixed effect. Panel A uses the bank's own ZLB Exposure, the twelve-quarter trailing average of its ZLB-era deposit inflows scaled by its total assets at the start of the episode (2008Q2 or 2019Q4). Panel B replaces it with the shift-share instrument ZLB Exposure^{Bartik} described in Table 7. Because this instrument is constructed from the bank's holdings of non-interest-bearing, savings, and time deposits at the start of the episode, it is mechanically related to deposit composition; Panel B therefore documents how the instrument loads on composition rather than providing independent evidence on the composition channels. QE Exposure$_{b,e}$ and ZLB Exposure are standardized within each episode and enter per standard deviation. $\ln(\text{Assets})_{b,e}$ is the log of total assets measured at the beginning of the three-year look-back window used to build the exposure measures. R^2 is adjusted. Standard errors, clustered by bank, are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Own ZLB Exposure

Dependent Variable:	(1) Uninsured Share	(2) Non-Int. Share	(3) Time Share
QE Exposure$_{b,e}$	0.019***	0.011***	0.003
	(0.002)	(0.002)	(0.002)
ZLB Exposure$_{b,e}$	0.046***	0.017***	-0.020***
	(0.002)	(0.002)	(0.002)
$\ln(\text{Assets})_{b,e}$	0.039***	-0.002	-0.025***
	(0.002)	(0.001)	(0.002)
N	10,074	10,075	10,075
R^2	0.320	0.060	0.177
Episode FE	Yes	Yes	Yes

Panel B: Shift-Share ZLB Exposure Instrument

Dependent Variable:	(1) Uninsured Share	(2) Non-Int. Share	(3) Time Share
QE Exposure$_{b,e}$	0.016***	0.006***	0.011***
	(0.002)	(0.002)	(0.002)
ZLB Exposure$^{\text{Bartik}}_{b,e}$	0.038***	0.060***	-0.098***
	(0.002)	(0.002)	(0.002)
$\ln(\text{Assets})_{b,e}$	0.035***	-0.010***	-0.010***
	(0.002)	(0.001)	(0.001)
N	9,960	9,960	9,960
R^2	0.308	0.272	0.528
Episode FE	Yes	Yes	Yes

Table A.7: Robustness of the ZLB and QE Effects to Small-Market Rate Gaps during Non-ZLB Periods

This table reproduces Table 7 using only "small" markets — counties in which the bank raises less than half of its total deposits. Because a bank sets its deposit rate based on conditions across all of its markets, the Rate Gap in the bank's dominant markets may be influenced by the bank's own choices; restricting to markets where the bank raises less than half of its deposits isolates the competing markets least likely to affect those choices, sharpening the exogeneity of the Rate Gap. Both panels pool the post-GFC and post-pandemic non-ZLB periods. Panel A uses bank-quarter Call Report data (2016Q1–2020Q1 and 2022Q2–2025Q4) and replaces the Rate Gap with one constructed only from the bank's small markets; the measure is defined only for banks operating in at least two counties. The dependent variable is the quarterly change in total (columns 1–2) or uninsured (columns 3–4) deposits, scaled by lagged assets; within each deposit category, the OLS column reports the endogenous ZLB $\text{Exposure}_{b,t-1}$ and the reduced-form column replaces it with its shift-share instrument $\text{ZLB Exposure}^{\text{Bartik}}_{b,t-1}$. Panel B uses bank-county-year Summary of Deposits (SOD) data (SOD years 2017–2020 and 2023–2025), restricted to bank-county observations in which the county held less than half of the bank's total deposits in the previous SOD year; the dependent variable is the annual change in bank b's deposits in county c, scaled by the bank's lagged total assets. Variable definitions follow Table 7; all exposure measures are standardized over non-ZLB periods and enter both directly and interacted with the Rate Gap. Panel A includes bank×episode and quarter fixed effects, with standard errors clustered two-way by bank and state-quarter. Panel B includes bank×episode fixed effects together with county×year (column 1) or county (column 2) fixed effects, with standard errors clustered two-way by bank and county. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Bank-Quarter, Small-Market Rate Gap

	Total Deposits		Uninsured Deposits	
	(1) OLS	(2) Reduced Form	(3) OLS	(4) Reduced Form
Dependent Variable:	$\Delta\text{Dep}_{b,t}/\text{Assets}_{b,t-1}$		$\Delta\text{Uninsured Dep}_{b,t}/\text{Assets}_{b,t-1}$	
$\text{Rate Gap}_{b,t}$	0.18**	0.38***	0.23***	0.42***
	(0.08)	(0.09)	(0.07)	(0.07)
$\text{ZLB Exposure}_{b,t-1}$	0.002***		0.001***	
	(0.000)		(0.000)	
$\text{ZLB Exposure}^{\text{Bartik}}_{b,t-1}$		-0.002***		-0.002***
		(0.000)		(0.000)
$\text{QE Exposure}_{b,t-1}$	0.000	0.000*	-0.001***	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)
$\text{Rate Gap}_{b,t} \times \text{ZLB Exposure}_{b,t-1}$	0.13***		-0.03	
	(0.04)		(0.03)	
$\text{Rate Gap}_{b,t} \times \text{ZLB Exposure}^{\text{Bartik}}_{b,t-1}$		0.17***		0.11***
		(0.05)		(0.04)
$\text{Rate Gap}_{b,t} \times \text{QE Exposure}_{b,t-1}$	0.28***	0.29***	0.15***	0.09***
	(0.04)	(0.04)	(0.03)	(0.03)
N	96,669	89,922	96,669	89,922
Bank×Episode FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes

Table A.7 continued from previous page

Panel B: Bank-County-Year, Small Markets		
	Total Deposits	
	(1)	(2)
Dependent Variable:	$\Delta \text{Dep}_{b,c,t} / \text{Assets}_{b,t-1}$	
$\text{Rate Gap}_{b,c,t}$	1.29***	0.86***
	(0.10)	(0.07)
$\text{ZLB Exposure}_{b,c,t-1}$	0.001*	-0.001**
	(0.000)	(0.000)
$\text{QE Exposure}_{b,c,t-1}$	0.001***	0.000
	(0.000)	(0.000)
$\text{Rate Gap}_{b,c,t} \times \text{ZLB Exposure}_{b,c,t-1}$	0.12*	0.19***
	(0.06)	(0.07)
$\text{Rate Gap}_{b,c,t} \times \text{QE Exposure}_{b,c,t-1}$	0.20***	0.21***
	(0.04)	(0.04)
N	81,717	84,965
Bank×Episode FE	Yes	Yes
County×Year FE	Yes	No
County FE	No	Yes

Table A.8: Deposit-Flow Sensitivity to the Rate Gap Excluding the March 2023 Banking Stress during Non-ZLB Periods

This table reproduces Table 7 excluding the period around the failure of Silicon Valley Bank, to verify that the results are not driven by the March 2023 banking stress. Both panels pool the post-GFC and post-pandemic non-ZLB periods. Panel A uses bank-quarter Call Report data (2016Q1–2020Q1 and 2022Q2–2025Q4), excluding 2023Q1–2023Q2; the dependent variable is the quarterly change in total (columns 1–2) or uninsured (columns 3–4) deposits, scaled by lagged assets. Within each deposit category, the OLS column reports the endogenous ZLB Exposure$_{b,t-1}$ and the reduced-form column replaces it with its shift-share instrument ZLB Exposure$^{\text{Bartik}}_{b,t-1}$. Panel B uses bank-county-year Summary of Deposits (SOD) data (SOD years 2017–2020 and 2023–2025), excluding SOD year 2023, which contains the SVB failure; the dependent variable is the annual change in bank b's deposits in county c, scaled by the bank's lagged total assets. Variable definitions follow Table 7; all exposure measures are standardized over non-ZLB periods and enter both directly and interacted with the Rate Gap. Panel A includes bank×episode and quarter fixed effects, with standard errors clustered two-way by bank and state-quarter. Panel B includes bank×episode fixed effects together with county×year (column 1) or county (column 2) fixed effects, with standard errors clustered two-way by bank and county. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Bank-Quarter, Excluding 2023Q1–2023Q2

	Total Deposits		Uninsured Deposits	
	(1) OLS	(2) Reduced Form	(3) OLS	(4) Reduced Form
Dependent Variable:	$\Delta\text{Dep}_{b,t}/\text{Assets}_{b,t-1}$		$\Delta\text{Uninsured Dep}_{b,t}/\text{Assets}_{b,t-1}$	
Rate Gap$_{b,t}$	0.45***	0.52***	0.33***	0.45***
	(0.08)	(0.09)	(0.06)	(0.07)
ZLB Exposure$_{b,t-1}$	0.003***		0.002***	
	(0.000)		(0.000)	
ZLB Exposure$^{\text{Bartik}}_{b,t-1}$		-0.001***		-0.001***
		(0.000)		(0.000)
QE Exposure$_{b,t-1}$	0.000	0.001**	-0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Rate Gap$_{b,t}$ × ZLB Exposure$_{b,t-1}$	0.05		-0.05	
	(0.04)		(0.03)	
Rate Gap$_{b,t}$ × ZLB Exposure$^{\text{Bartik}}_{b,t-1}$		0.05		0.12***
		(0.05)		(0.04)
Rate Gap$_{b,t}$ × QE Exposure$_{b,t-1}$	0.25***	0.25***	0.14***	0.11***
	(0.03)	(0.03)	(0.02)	(0.02)
N	151,343	139,302	151,341	139,302
Bank×Episode FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes

Table A.8 continued from previous page

Panel B: Bank-County-Year, Excluding SOD Year 2023		
	Total Deposits	
	(1)	(2)
Dependent Variable:	$\Delta \text{Dep}_{b,c,t} / \text{Assets}_{b,t-1}$	
$\text{Rate Gap}_{b,c,t}$	3.25***	2.16***
	(0.21)	(0.15)
$\text{ZLB Exposure}_{b,c,t-1}$	0.001**	-0.001***
	(0.000)	(0.000)
$\text{QE Exposure}_{b,c,t-1}$	0.003***	0.002***
	(0.000)	(0.000)
$\text{Rate Gap}_{b,c,t} \times \text{ZLB Exposure}_{b,c,t-1}$	0.11**	0.17***
	(0.04)	(0.04)
$\text{Rate Gap}_{b,c,t} \times \text{QE Exposure}_{b,c,t-1}$	0.22***	0.25***
	(0.06)	(0.06)
N	103,053	104,210
Bank×Episode FE	Yes	Yes
County×Year FE	Yes	No
County FE	No	Yes